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# Multi-agent reinforcement learning for locally observable cooperative systems with acyclic dependence structure

Claire Bizon Monroc <sup>\*</sup>, Ana Bušić <sup>†</sup>, Donatien Dubuc <sup>,</sup>, and Jiamin Zhu <sup>‡</sup>

**Abstract.** Single-agent reinforcement learning algorithms can be directly applied to multi-agent systems in an independent learning approach, but often lose their convergence properties as the environment becomes non-stationary. We prove that in transition-independent Decentralized Markov Decision Process (TI Dec-MDP) non-stationarity can be mitigated by a multi-scale approach when the inter-dependence of agents' dynamics can be represented by a directed acyclic graph (DAG). Although TI Dec-MDPs are thought to be restrictive, we show that they can describe complex real world problems through a careful design of local states. We propose a multi-scale Q-learning algorithm (MQL) where agents update local Q-learning iterates at different timescales without communication and still converge. To this purpose, we first show that we can model the loss of information on the global state as a state-dependent Markovian noise. Then, we show that results from stochastic approximation theory can be used to prove the convergence of MQL under local state observability. We give practical solutions to exploit knowledge about agent interaction to assign learning rates that ensure convergence, and propose a NetworkMQL algorithm that can achieve convergence in Network-Distributed POMDP (ND-POMDP). We validate both MQL and NetworkMQL on a wind farm control problem.

**Key words.** Learning and adaptive systems in artificial intelligence, Online algorithms, Multi-agent systems, Decentralized systems

**1. Introduction.** Recent advances in reinforcement learning (RL) have seen a growing interest in solving cooperative multi-agent problems, where several agents interact with the same environment to optimize a common objective [44, 29]. Multi-agent reinforcement learning (MARL) has encountered successes in fields as varied as games with multiple players [5], vehicle routing problem for traffic regulation [43], or distributed optimal control of wind farms [36, 9, 20]. In this article, we consider a family of fully cooperative, infinite horizon multi-agent reinforcement problems, where state information is distributed among all agents and they must collaborate to maximize a shared reward. Such problems are commonly formulated as decentralized partially observable Markov decision processes (Dec-POMDPs) in the MARL literature. It is known that solving Dec-POMDPs is very hard: finding a solution to a two-player Dec-POMDP has been proven to be NEXP-hard, and is undecidable for the infinite-horizon case [8]. Instead of general Dec-POMDP, we focus on the special case of the *transition-independent* Dec-MDP, which is of NP-hard complexity [3]. In a transition-independent Dec-MDP, each agent's local observations only depend on local states and actions, so that agents only interact through the shared reward. In general, blind cooperation problems in which agents must learn to coordinate while being oblivious to each other's existence will fit this description. In the rover exploration problem introduced by [7] for example, several rovers must coordinate to explore a planet. Rovers are assigned distinct sides of the planet to explore

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so that they do not directly interact, but the value of the information they can gather depends on what is collected by other agents.

Transition-independent Dec-MDPs have been studied in the planning literature, and several planning algorithms have been proposed to find optimal local policies [28, 7, 17, 14]. However, they exploit full knowledge of the transition matrix and the reward function, which are rarely available in real-life problems. MARL algorithms, on the other hand, focus on learning solely from interactions with the environment. In a centralized training, a single learner updates all local policies based on global observation. Multi-agent systems are, however, often under constraints that prevent instantaneous communication of all agents with a central controller. Moreover, centralized approaches are vulnerable to the curse of dimensionality, as the size of the search space increases exponentially with the number of agents.

In an online decentralized training approach, each agent updates its own local policy only with the information it has collected. To compensate for the loss of information, online learning algorithms for Dec-POMDP typically allow agents to keep past observations in their memory, allow communication between agents during training, maintain beliefs about the global state of the environment, or estimate the behaviors of other agents [28, 29]. Instead, an interesting question is to see whether memory-less policies can still be learned without additional estimates or communication. In particular, MARL has focused on extending the success of single-agent reinforcement learning algorithms to the multi-agent case, such as the classic Q-learning algorithm [41]. One naive way to adapt the single-agent Q-learning to the multi-agent case is to simply let each agent run local Q-learning updates, with other agents considered as a part of its environment. This approach was first introduced by [39] as *Independent learning*. Independent learning has been shown to produce good empirical results in a number of problems [24], and is used as a baseline to develop current state-of-the-art MARL algorithms [42]. As it avoids the exponential dependence of dimension of the search space in the number of agents, independent learning is more scalable. However, agents learning simultaneously make the state and reward processes appear non-stationary for each agent, an issue that is commonly referred to as the *non-stationarity* problem. This leads to the loss of convergence guarantee for single-agent RL algorithms. Moreover, the system may prevent full observability of the environment to the agents, with each agent only collecting a function of the global state. Solutions to mitigate the non-stationarity typically rely on a modification of algorithm's local rule. Examples have been turn-based control [32] or turn-based updates [37], two stages algorithms requiring agent coordination during learning [34, 4], coordinated exploration between agents, and consensus-based approaches or explicit modeling of other agent's behavior [44]. We propose a multi-scale approach that does not require coordination between agents during learning, and converges to a best-response equilibrium in transition-independent Dec-MDP.

Although transition-independent Dec-MDPs are thought to be restrictive, our first contribution is to show that they can be imposed *by design* to describe complex real world problems, showing a trade-off between observability and complexity. In this article, we show that transition-independent Dec-MDPs can be constructed from certain standard Dec-POMDPs, and that such a reformulation can be useful to solve real industrial problems. In [25] for example, a distributed wind farm optimization problem is considered, in which the local information of the Dec-POMDP can be factorized into two components. The first is a private

component, that is a local component independent of other agents. The second is a deterministic function of the private components of other agents, and of an exogenous Markovian process that is independent of any agent’s action. Whenever such an exogenous process can be identified, constructing local states by replacing the second component with a direct observation of the exogenous process frees the local state from dependence on other agents’ action, while maintaining the Markovian property of the global MDP. A detailed example will be given in Section 4. This reformulation could explain the good empirical results of the independent learning approaches applied to wind farm control problems observed in the literature [25, 20] (see [2] for a thorough review).

We show that for independent learners in transition-independent Dec-MDP, the loss of information due to partial observability can be seen as a Markovian noise. We then focus on the case where agent dynamics can be described by a directed acyclic graph (DAG), and show that in such a case the non stationarity issue can be addressed by a multi-timescale learning approach: by allowing agents to learn at different time scale, we ensure that for any “fast” agent, the “slow” evolution of other policies will lead iterates to behave as though the environment was stationary, and ultimately converge. Note that this approach belongs to the independent learning family and thus is scalable to large number of agents. Moreover, we show that our multi-scale Q-learning approach can further exploit information on interactions between agents to accelerate convergence. For this purpose, we focus on the network distributed POMDPs (ND-POMDPs), a class of transition independent Dec-POMDP introduced in [26], where agent interactions can be represented by a sparse graph. [26] focuses on the finite-horizon problem where the dynamics of the MDP are fully known, and proposes a decentralized planning algorithm relying on communication between agents and costly best response computations at each step. In our reinforcement learning framework, the environment dynamics are however unknown, the horizon is infinite and learning must be done online: we show how our independent learners still converge to a best response equilibrium when the graph is acyclic.

Our convergence analysis is based on stochastic approximation techniques. Stochastic approximation and multi-timescale approaches have been used to analyze fully observable learning algorithms. In [11], the convergence of the single-agent Q-learning has been proven with a stochastic approximation approach. Two-timescale stochastic approximation was then introduced in [10], showing that two interdependent stochastic processes can both converge when they are updated at two different scales. These have been successfully used to build reinforcement learning algorithms maintaining different iterates. The learning of future rewards and of the best response has for example been decoupled in various fictitious-play [6, 33, 23] and Q-learning [31] inspired schemes for fully observable zero-sum and team games. Two-timescale results have been extended to an arbitrary number of scales in [22], and used to analyze a multi-scale learning algorithm in some classes of repeated games. In [27], a similar multi-scale approach is further evaluated on several multi-agent reinforcement learning problems, but a convergence analysis is not provided. Unfortunately, convergence results of these works have required the reward function to be stationary, meaning that for a given agent observation-action pair, the collected reward is always sampled from the same distribution. These results therefore cannot apply to the Dec-MDP case where global state information is distributed among agents, and each can only observe a local state. We make the following

contributions.

- We show that for independent learners in a transition-independent Dec-MDP, the loss of information due to partial observability can be seen as a Markovian noise.
- We prove that in a class of Dec-MDP where agent dynamics can be described by a DAG, independent Q-learners learning at different time scales can converge towards a best-response equilibrium. To achieve this, we first formally extend the weak convergence [Theorem 6.2, Chapter 8, [21]] for synchronous two-timescale iterates to synchronous multi-timescale iterates, then further extend this result together with [Theorem 5.1, Chapter 12, [21]] for single-timescale asynchronous updates to asynchronous multi-timescale updates (see Theorem 5.1)
- We show that in ND-POMDPs, our multi-scale Q-learning approach can exploit a DAG structure of interactions to accelerate convergence.
- We apply these results to the maximization of power output in the wind farm control problem. We show that it can be rewritten as a transition-independent Dec-MDP, providing a first explanation for the success of independent learners on this task observed in the literature.
- We show that our DAG structure can be used to design local rewards and accelerate convergence in the wind farm control problem.

The paper is organized as follows. In Section 2 we formalize the problem of finding an equilibrium in a transition-independent Dec-MDP. Then in Section 3, we propose a multi-scale Q-learning algorithm: we lay out the assumption of acyclic dependence structure between agent dynamics, and show how it allows us to apply our multi-scale algorithm to TI Dec-MDPs. We then exploit the graph of interaction between agents in a networked problem to derive faster algorithms. Our experiment in Section 4 evaluates the multi-scale approach on the real industrial problem of wind farm control, and empirically validates its convergence. The last section of this paper is then dedicated to proofs: in Section 5, we show that the convergence of our multi-scale Q-learning can be derived from the weak convergence of multi-scale iterates with Markovian noise. The formal extension of the two-timescale convergence results introduced in [21] to an arbitrary number of scales with asynchronous updates is detailed in the supplement.

**2. Cooperative MARL with local learners.** We start by formalizing the problem of transition-independent Dec-MDP. We then explicit the assumptions on the transitions and local policies that we will consider in the rest of this paper, before introducing our multi-scale Q-learning algorithm.

**2.1. Independent transition Dec-POMDP.** We consider a decentralized partially observable Markov Decision Process (Dec-POMDP) reinforcement learning problem, where  $M$  agents interact with the same environment to maximize a common reward. Let us assume a finite state space  $S$  and a finite action space  $A$ . The global action space  $A$  is factorized into  $M$  local action spaces  $A = A_1 \times \dots \times A_M$ . Any agent  $i$  only receives a partial observation of the global state  $o^i$ , and samples an action according to a local policy  $a^i \sim \pi^i$  mapping observations to distributions over  $A_i$ . This induces a global policy  $\pi = (\pi^1, \dots, \pi^M)$ . A global reward  $r : S \times A \rightarrow \mathbb{R}$  is shared by all agents. The reward is bounded in  $\mathbb{R}$  by a constant  $R > 0$ , that is:  $\forall (s, a) \in S \times A, |r(s, a)| \leq R$ . We write  $P : S \times A \times S \rightarrow (0, 1)$  a transition kernel,

denoting transition probabilities between states given chosen actions. For a discount factor  $\beta \in (0, 1)$ , we wish to maximize the expectation of the sum of discounted reward or return  $\mathbb{E}_\pi [\sum_{k=0}^{\infty} \beta^k r(s_k, a_k)]$ . As we consider a transition-independent Dec-POMDP [7], we assume that transitions between locally observed states only depend on local states and actions:

A 2.1.  *$S$  is factored into local state spaces  $S = S_1 \times \dots \times S_M$  and there exists local transition kernels  $(P^i)_{i=1\dots M}$  such that  $\forall s, a, s' \in S \times A \times S$ ,  $s = (s^1, \dots, s^M)$  with:*

$$P(s, a, s') = \prod_{i=1}^M P^i(s^i, a^i, s'^i)$$

If the partial observation  $o^i$  received by agent  $i$  is exactly its local state  $s^i$ , then the transition-independent Dec-POMDP is fully locally observable and is also called a transition-independent decentralized MDP (TI Dec-MDP).

This model can describe several problems in robotics or industrial applications. The collective rover exploration problem introduced in [7] for example considers rovers with independent dynamics and a shared reward. Note that the model can be extended to a factorization of the state space in  $M + 1$  components  $S = S_e \times S_1 \times \dots \times S_M$ , where  $S_e$  is the state space of an exogenous Markovian process. An exogenous process is not affected by any agent's actions. The transition of that exogenous process is then defined by a local kernel  $P(s^e, s'^e)$  such that  $P(s, a, s') = P(s^e, s'^e) \prod_{i=1}^M P^i(s^i, a^i, s'^i)$ . The  $i$ th agent can receive the partial observation  $(s^e, s^i)$ . In Section 4, we will consider the problem of the cooperative control of many wind turbines in a wind farm, and show how it can be rewritten as a TI Dec-MDP with an exogenous process.

It has been shown by [16] and [13] that in TI Dec-MDPs the return can be maximized by decentralized Markovian policies: policies that only depend on the last observed local state. Denoting for all  $s$  and  $a$ ,  $\pi(a | s)$  the probability of action  $a$  (the vector of individual actions of all agents) in state  $s$ , we have  $\pi(a | s) = \prod_{i=1}^M \pi^i(a^i | s^i)$ .

For any stationary global policy  $\pi$ , the global state process  $\mathbf{s}$  is in fact a Markov chain with transition matrix

$$P_\pi(s, s') = \sum_a \pi(a | s) P(s, a, s') = \sum_{a=(a^1, \dots, a^M)} \prod_{i=1}^M \pi^i(a^i | s^i) P(s, a, s')$$

We now introduce an assumption on the transition function of the MDP

A 2.2. *For any non-deterministic local policy  $\pi^i$  such that  $\forall a^i, s^i, \pi^i(a^i | s^i) > 0$ , the local state process is an irreducible and aperiodic Markov chain.*

This classical assumption for Q-learning [41, 40, 19] will ensure that all local state processes admit an invariant distribution, and will converge to it under a fixed policy regardless of the initial distribution. Note that following A 2.1 this implies that the global state process is also irreducible and aperiodic.

Using vector notation, we define  $d^\pi \in (0, 1)^{|S|}$  the invariant distribution over the global state space  $S$  satisfying  $d^\pi P_\pi = d^\pi$ . Similarly, for every agent  $i$  we define  $d_i^{\pi^i}$ , the invariant distribution of the local state process  $\mathbf{s}^i$ . If we ensure that local policies  $\pi^i$  have non-null probabilities on all the local action space, then A 2.2 implies that the local state-action process  $(\mathbf{s}^i, \mathbf{a}^i)$  is also irreducible: it is a Markov chain over  $S_i \times A_i$ , with transition matrix

203  $P_{\pi^i}((s^i, a^i), (s'^i, a'^i)) = P(s^i, a^i, s'^i) \pi^i(a'^i | s'^i)$  given by A 2.1. We denote its invariant distribu-  
 204 tion as  $\lambda^i$ .

205 In the rest of this paper, we consider the TI Dec-MDP that satisfies A 2.1 and A 2.2.  
 206 For simplicity of notations, in the following we ignore exogenous processes, but the analysis  
 207 is easily adapted to them.

208 **2.2. Q-functions in a TI Dec-MDP.** For an agent  $i$  and a global policy  $\pi$ , we note  $\pi^{-i}$   
 209 the set of local policies in  $\pi$  except  $\pi^i$ . For any pair  $(s^i, a^i) \in S_i \times A_i$  and any global policy  $\pi$ ,  
 210 we define the  $i^{th}$  q-function  $Q_{\pi^i}^{\pi^{-i}}(s^i, a^i)$  the value of taking action  $a^i$  in local state  $s^i$ , and then  
 211 following policy  $\pi^i$ , provided that any other agent  $j$  follows its respective local policy  $\pi^j$ . As  
 212 the reward is a function of the global state, the value expectation must be taken with respect  
 213 to a distribution on the global state. Noting that we consider the infinite-horizon problem, we  
 214 chose the stationary distribution  $d^\pi$ , to which the global state process asymptotically converges  
 215 under  $\pi$ , under assumption A 2.2. A similar extension of the value function has been proposed  
 216 for the single-agent partially observable case (POMDP) [35]. We therefore have:

$$217 \quad (2.1) \quad Q_{\pi^i}^{\pi^{-i}}(s^i, a^i) = \mathbb{E}_{s_0 \sim d^\pi, a_k \sim (\pi^i, \pi^{-i}), s_k \sim P} \left[ \sum_{k=0}^{\infty} \beta^k r(s_k, a_k) \mid s_0^i = s^i, a_0^i = a^i \right]$$

218 These local q-functions  $Q_{\pi^i}^{\pi^{-i}}$  can be written as tables of dimension  $|S_i| \times |A_i|$ , and admit a  
 219 recursive formula given in Lemma 2.1. Note that this recursive formula exists because the  
 220 value expectation is taken with regard to the stationary distribution on the global state.

221 **Lemma 2.1.** *Any local q-function (2.1) satisfies the following recursive formula:*

$$222 \quad (2.2) \quad Q_{\pi^i}^{\pi^{-i}}(s^i, a^i) = \sum_s d^\pi(s \mid s^i) \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid s) r(s, a) + \beta \sum_{s'^i} P^i(s^i, a^i, s'^i) \sum_{a'^i} \pi^i(a'^i \mid s'^i) Q_{\pi^i}^{\pi^{-i}}(s'^i, a'^i)$$

223 The proof of Lemma 2.1 is straightforward but tedious, we detail it in the Supplement (Sec-  
 224 tion 1). Like for the single-agent q-value function [38], the q-value is divided into two parts:  
 225 an immediate reward collected in the current state and a future gain, that is, the reward  
 226 expectation starting from the next state. We now introduce the definition of a best response,  
 227 as a local policy  $\pi^i$  that maximizes the return when other local policies are fixed.

**Definition 2.2 (Best response).** *A local policy  $\pi_{br}^i$  is said to be a best response to a set of local policies  $\pi^{-i}$  if starting from any local state, it always maximizes the return as the other agents follow local policies  $\pi^{-i}$ . That is, for any local policy  $\pi^i$  we have:*

$$Q_{\pi_{br}^i}^{\pi^{-i}}(s^i, a^i) \geq Q_{\pi^i}^{\pi^{-i}}(s^i, a^i) \quad \forall s^i \in S_i, a^i \in A_i$$

228 Best response policies will therefore maximize the expectation of this optimal q-value  
 229 at every state  $s^i$ . They can be written as the set of policies  $\pi^i$  such that  $\pi^i(\cdot \mid s^i) \in$   
 230  $\arg \max_{\rho \in \Omega(s^i)} [\rho^T Q_{\pi^i}^{\pi^{-i}}(s^i, \cdot)]$ , where  $\Omega(s^i) \subset [0, 1]^{|A_i|}$  is the simplex of dimension  $|A_i|$  represent-  
 231 ing the set of local strategies mapping a given local state to a distribution over actions. Yet

in order to ensure that local policies always have non-null probabilities on the local action space, we consider a regularized objective introduced in [22, 31]: for any given q-value table  $Q^i$ , let us define the mapping  $\phi$  that returns the following local policy:

$$(2.3) \quad \phi(Q^i)(\cdot | s^i) = \arg \max_{\rho \in \Omega(s^i)} [\rho^T Q^i(s^i, \cdot) + \tau \nu_{s^i}^i(\rho)] \quad \forall s^i \in S_i$$

where  $\tau > 0$  is a temperature parameter representing the weight given the regularization, and  $\nu_{s^i}^i$  is a smooth and strongly concave function which takes infinite values outside of  $\Omega(s^i)$ . Strong concavity ensures the uniqueness of the solution  $\phi(Q^i)$ , and we call any local policy  $\pi^{*i}$  such that  $\pi^{*i} = \phi(Q_{\pi^{*i}}^{\pi^{*i}})$  a smoothed best response to  $\pi^{*i}$ . If all agents follow a smoothed best-response, then the corresponding global policy is called an equilibrium.

**Definition 2.3 (Equilibrium).** *A global policy  $\pi^*$  is an equilibrium iff every local policy  $\pi^i$  is a smoothed best response to other local policies  $\pi^{-i}$ .*

To shorten the notation, we write  $v'(Q^i, s^i, a^i)$  the expectation of the future gain as estimated by any table  $Q^i$  after taking action  $a^i$  in  $s^i$ :

$$(2.4) \quad v'(Q^i, s^i, a^i) = \sum_{s'^i} P^i(s^i, a^i, s'^i) [\phi(Q^i)(\cdot | s'^i)]^T Q^i(s'^i, \cdot)$$

Then writing  $Q_*^{\pi^{*-i}} = Q_{\pi^{*-i}}^{\pi^{*-i}}$ , from Lemma 2.1 the equilibrium  $\pi^*$  and its associated q-functions  $Q_*^{\pi^{*-i}}(s^i, a^i)$  are solutions to the following equations: for all  $i \in \{1 \dots M\}$ ,  $s^i \in S_i$ ,  $a^i \in A_i$ ,

$$Q_*^{\pi^{*-i}} = \sum_s d^{\pi^*}(s | s^i) \sum_{a^{-i}} \pi^{*-i}(s, a^{-i}, a^{-i}) + \beta v'(Q_*^{\pi^{*-i}}, s^i, a^i)$$

**3. Weak-convergence of multi-scale Q-learning iterates.** Let all agents maintain a local estimate  $\hat{Q}^i$  of the q-function (2.1), and follow a local policy  $\pi^i = \phi(\hat{Q}^i)$ . The combined actions of all agents sample  $M$  local trajectories  $\{(s_0^i, a_0^i, r_0^i), (s_1^i, a_1^i, r_1^i) \dots\}$ ,  $i \in \{1 \dots M\}$ . Let now all agents locally run a Q-learning update, so that each agent updates its local estimate  $\hat{Q}_k^i$  at each timestep  $k$ :

$$(3.1) \quad \begin{aligned} \hat{Q}_{k+1}^i(s^i, a^i) &= \hat{Q}_k^i(s^i, a^i) \\ &+ \alpha_k^i(s_k^i, a_k^i) \left[ r_k + \beta [\phi(\hat{Q}_k)(\cdot | s_{k+1}^i)]^T \hat{Q}_k(s_{k+1}^i, \cdot) - \hat{Q}_k(s_k^i, a_k^i) \right] I_{k, s^i, a^i} \end{aligned}$$

with  $I_{k, s^i, a^i}$  the indicator of the event that the local state-action pair  $s^i, a^i$  is visited at timestep  $k$ , and  $\alpha_k^i(s_k^i, a_k^i)$  learning rate sequences maintained separately by each agent. At this timestep, all other state-action pairs are not updated, and the iterates are therefore asynchronous.

Note that in single-agent Q-learning, the collected reward  $r(s, a)$  is exactly the expectation of the reward for the observed state-action pair  $(s, a)$ . Here however, no agent ever collects a reward sampled according to the stationary distribution of the equilibrium policy as defined in the q-value (2.1). In fact, no agent ever collects a reward sampled from any fixed distribution at all, as the reward is a function of the changing actions of other agents. Instead, we show that the collected reward depends on an unobserved Markovian global state process, and that

the difference between the collected reward and the reward expected at equilibrium can be seen as a state-dependent noise. To treat this state-dependent noise, we apply techniques from the stochastic approximation theory. We reframe the convergence of our Q-learning iterates as the problem of finding a specific ranking among agents, such that at equilibrium the change in policy of an agent of higher rank creates only small perturbations among agents with a lower rank. Then, we let agents of different ranks learn at different time scales by imposing a hierarchy between their respective learning rates  $\alpha_k^i$ . We call the resulting algorithm the *multi-scale Q-learning* algorithm.

In the next section we detail further what we mean by agent ranking. To simplify the exposition, in Subsection 3.1, we will assume that there is a total ordering of agents satisfying A 3.1. Intuitively, we assume that there is a total ordering of agents such that for any agent  $i$ , any small perturbation of the policy  $\pi^i$ , the best response dynamics of agents that are greater than  $i$  leads to a small change in the overall objective function. However, a total ordering will force us to have as many learning rates as agents. In Subsection 3.2, we will relax this assumption to a more realistic partial ordering of agents, and illustrate how to assign learning rates to different agents in a more concrete application. We will consider a networked distributed POMDP (ND-POMDP), in which the shared reward is distributed among agents, and the dependency graph between agents is acyclic and known. We show how the knowledge about this graph can be exploited to reduce the number of different learning rates and build a faster algorithm.

**3.1. Interaction structure between agents for multi-scale Q-learning.** Consider the case where each agent  $i$  is updating a local q-value estimate  $\hat{Q}^i$ , and the agents are given learning rates such that each agent is learning at a different timescale. We start by defining precisely the total preorder needed on agents for iterates (3.1) to converge.

Consider any agent  $i$  with a q-table  $Q^i$ , and write

$$(3.2) \quad \pi = (\pi^1, \dots, \pi^M), \quad \pi^j = \phi(Q^j), \quad Q = (Q^1, \dots, Q^M)$$

We also denote  $Q^{>i} = (Q^{i+1}, \dots, Q^M)$  and  $Q^{<i} = (Q^1, \dots, Q^{i-1})$ . Let us take a set of q-tables  $Q$  with its corresponding global policy  $\pi = \phi(Q)$ . According to A 2.2, there exists a corresponding stationary distribution over the global states  $d^\pi$ . We write  $r_\pi$  the expectation of the reward under  $d^\pi$ , so that  $r_\pi = \sum_s \sum_a d^\pi(s) \pi(a | s) r(s, a)$ . Now let us assume that:

- For  $j \leq i$ ,  $Q^j$  is any q-table in  $|S_j| \times |A_j|$
- For  $j > i$ ,  $Q^j$  is a q-table of the smoothed best response to  $\pi^{-j}$  defined in Subsection 2.2, and written  $Q^j = Q_{\pi^{-j}}^{\pi^{-j}}$ . We write  $Q^{>i} = Z^{\geq i+1}(Q^{<i}, Q^i)$ , where  $Z^{\geq i+1}$  returns the set of  $M - i$  best-response q-values.

Any disturbance to a local q-table  $Q^i \neq Q^i$  causes a corresponding change to  $Q^{>i}$ . If a local perturbation produces a change in the reward expectation smaller than the perturbation, then it will follow that the dynamics of the local iterates (3.1) can be approximated by an ODE with a unique fixed point. We now formalize this condition.

A 3.1. Let  $Q'^i \in [-D, D]^{|S_i| \times |A_i|}$  be a local perturbation to  $Q^i$  within the constraint set. Write  $Q' = (Q^{<i}, Q'^i, Z^{\geq i+1}(Q^{<i}, Q'^i))$  and  $\pi' = \phi(Q')$ . There exists  $K \in (0, 1)$  such that for every agent  $i$  and its q-table  $Q^i$ , the reward function satisfies:

$$|r_\pi - r_{\pi'}| \leq K \|Q^i - Q'^i\|_\infty$$

This assumption defines a specific contractive behavior of the expected reward after best response adaptation. It means that for every agent  $i$ , there exists a subset of other agents for which a small perturbation of  $i$  triggers a smaller deviation in the average reward. This will allow us to show that an ODE approximating our iterates converges towards a fixed point. When  $\phi$  returns the Boltzmann distribution, this contraction property can be understood as restricting the variation of the reward over the state-action space. A 3.1 is then verified when the maximum change in  $q$ -values for all local best-responses is not greater than the change in  $Q^i$ . To build intuition, we derive a bound over reward variation for a toy example in the Supplement (Section 3), and then extend it to the general case. As A 3.1 moreover defines a total order over all agents, we can let each agent learn at a different time-scale depending on its rank.

The choice of the  $M$  learning rate sequences  $\alpha_k^i$  will then play an important part in the convergence of iterates (3.1). In particular we will require two sets of assumptions. First, standard assumptions on the convergence of stochastic approximation iterates will require the learning rate sequences to both converge towards 0 and evolve relatively slowly. Recall that all state-action pairs are not updated at each iteration, and that the iterates are therefore asynchronous: we will therefore constraint the evolution of the sequences so that at any time-step learning rates for different local state-action pairs are not too different from each other. A second set of assumptions will ensure that each agent learns at a different time scale. In particular, for each  $i < j$ , we will have  $\frac{\alpha_k^i}{\alpha_k^j} \rightarrow 0$ , as  $k \rightarrow \infty$ , where  $i < j$  reflects the relationship in the order imposed by A 3.1. For now, we refer to these assumptions as A 5.5 and will give their formal introduction in Subsection 5.1. In Section 4, we construct a set of learning rate sequences that approximately satisfies these assumptions and show that this will be sufficient to lead to convergence in experiments inspired by a real-world problem. We now state our convergence theorem for the multi-scale Q-learning algorithm.

**Theorem 3.1.** *Let us consider  $M$  agents locally updating their  $q$ -value estimates according to (3.1) with initial values  $\hat{Q}_0^i \in [-D, D]$  for  $D > 0$  such that  $D > \frac{R}{1-\beta}$ . Suppose that A 3.1 is satisfied with the ordering of agents  $\{1, \dots, M\}$ , and the learning rates  $\{\alpha_i\}_{1 \dots M}$  follow A 5.5, where  $\alpha_i$  is the learning rate sequence of the  $i^{\text{th}}$  agent. If the discount factor  $\beta$  satisfies  $\beta \leq 1 - K$ , then the  $q$ -value estimates will converge weakly towards the smoothed best-response  $q$ -values  $Q^{*i}$ . Moreover, the global policy defined by  $\pi^{*i} = \phi(Q^{*i})$  for all  $i$  is an equilibrium.*

The proof of this theorem will be detailed in Subsection 5.2. It requires every agent to learn at a different time-scale. However, convergence can be preserved in other cases where the dependency between agent dynamics is acyclic. This dependency can then be represented by a directed acyclic graph (DAG). If this graph is known, it can be used to assign a ranking to agents that allows different agents to have the same learning rate sequence. In the next section we look at a class of Dec-POMDP with specific assumptions on agent interaction structure and see how this allows us to reduce the number of ranks and derive a faster algorithm.

**3.2. Reward decomposition for multi-scale Q-learning.** We now consider the Networked Distributed POMDPs (ND-POMDPs), a specific case of transition-independent Dec-POMDPs introduced by [26] to model distributed optimization problems. We now assume the shared reward can be written as a sum of  $M$  components  $\{r^i\}_{1, \dots, M}$  such that for all  $(s, a) \in S \times A$ ,

$r(s, a) = \sum_{i=1}^M r^i(s^i, a^i, s^{U^i}, a^{U^i})$ , where  $U^i$  is a subset of agents, and  $s^{U^i}$  - resp.  $a^{U^i}$  - is a vector concatenating local states - resp. local actions - of agents in  $U^i$ . We say that agent  $i$  influences agent  $j$  if  $i \in U^j$ . Here, the total reward is not received by every agent, but rather distributed in the network that connects all agents.

Let the relationships between agents be modeled by a directed acyclic graph (DAG)  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  with  $\mathcal{V}$  the set of vertices and  $\mathcal{E}$  the set of edges, such that  $|\mathcal{V}| = M$ , and  $(i \rightarrow j) \in \mathcal{E}$  iff agent  $i$  influences agent  $j$ . For every node  $i$ , we write  $\mathcal{N}_{in}^i$  the set of nodes from which there is an edge to  $i$  in the graph, and  $\mathcal{N}_{out}^i$  the set of nodes to which there is an edge from  $i$  in the graph. The neighborhood of node  $i$  is then noted  $\mathcal{N}^i = \mathcal{N}_{in}^i \cup \mathcal{N}_{out}^i$ . We write  $\mathcal{NA}(i)$  the ancestors of  $i$ , that is the set of nodes for which there exists a path towards  $i$ . Similarly, we write  $\mathcal{ND}(i)$  the descendants of  $i$ . Under A 5.5, every agent learned at a different scale, for a total of  $M$  different scales. In ND-POMDPs, we can exploit the structure of the problem to attribute a smaller set of  $\bar{M} \leq M$  scales to all agents.

We want to find a ranking function  $rk : i \rightarrow rk(i) \in \{1, \dots, \bar{M}\}$ , such that the convergence of Theorem 3.1 is preserved if every agent  $i$  is assigned the learning rate sequence  $\alpha_k^{rk(i)}$ . Let us start by rewriting A 3.1 as a looser, local assumption A 3.2. In the general case, the dynamics of all iterates were assumed to be dependent on all other iterates, and A 3.1 therefore imposed a total order on the dynamics of all agents. Under our DAG structure however, if the parents of agent  $i$  are fixed, then only a - possibly strict - subset of other agents need to adapt their best responses to a change in its policy. We write  $Q^S$  the q-tables of any subset of agents  $S \subset \mathcal{V}$ , so that  $Q = (Q^{\mathcal{V} \setminus (i, \mathcal{ND}(i))}, Q^i, Q^{\mathcal{ND}(i)})$ , and similarly we write the set of best-response functions  $Z^S$ . If  $Q^i$  is the local perturbation of the table  $Q^i$ , we look at the behavior of  $Q'$  when  $Q^{\mathcal{V} \setminus (i, \mathcal{ND}(i))}$  remains fixed. Indeed, agents in  $\mathcal{V} \setminus \mathcal{ND}(i)$  either need not change their policies because a change in  $i$  does not change their best responses, or can appear "fixed" at  $i$ 's scale by learning at a slower scale than  $i$ . The possibility to gain in learning speed will depend on the size of the first subset.

**A 3.2.** *Let  $Q' = (Q^{\mathcal{V} \setminus (i, \mathcal{ND}(i))}, Q^i, Z^{\mathcal{ND}(i)}(Q^i, Q^{\mathcal{V} \setminus (i, \mathcal{ND}(i))}))$ . For every agent  $i$  in  $\mathcal{G}$  and with the same notations as A 3.1, there exists  $K \in (0, 1)$  such that  $|r_\pi - r_{\pi'}| \leq K \|Q^i - Q'^i\|_\infty$*

Then, any ranking that satisfies the following conditions preserves the convergence of Theorem 3.1

(A) For any node  $i$ , nodes in  $\mathcal{NA}(i)$  have a strictly inferior rank, and nodes in  $\mathcal{ND}(i)$  have a strictly superior rank.

(B) For any node  $i$ , there exists no two different nodes of the same rank in  $\mathcal{NA}(i)$ .

Rankings that satisfy (A) and (B) exist. For example, if we apply any topological sorting algorithm to our DAG, it return  $\bar{M} = M$  ranks: the total order will satisfy (A) by construction, and trivially satisfy (B) by giving a different rank to every node. As another example, if  $\mathcal{G}$  is a tree, finding a learning rate attribution that satisfies (A) and (B) is immediate: we only need to give to every node the size of the longest path from the root - a node without incoming edge - to this node. This remains true if we add to  $\mathcal{G}$  any edge going from a node to one of its descendants. We give in the Supplement (Algorithm 4.1, Subsection 4.1) a straightforward attribution procedure for any DAG that returns  $\bar{M} < M$  ranks whenever it is possible. An example of the application of that procedure to a real example will be introduced in Section 4. We now state the extension of Theorem 3.1.

Theorem 3.2. Consider  $M$  agents locally updating their  $q$ -values estimates according to iterates

$$(3.3) \quad \hat{Q}_{k+1,c}^i = \hat{Q}_{k,c}^i + \alpha_k^i(s_k^i, a_k^i) \left[ \bar{r}_k^i + \beta[\phi(\hat{Q}_k^i)(s_{k+1}^i)]^T \hat{Q}_k^i(s_{k+1}^i, \cdot) - \hat{Q}_{k,c}^i \right]$$

with  $\bar{r}_k^i = \sum_{j \in \{i, \mathcal{N}^i\}} r^j(s_k^j, a_k^j, s_k^{U^j}, a_k^{U^j})$ , in the ND-POMDP with DAG interaction network  $\mathcal{G}$  satisfying A 3.2. Let the learning rates  $\{\alpha_k\}$  be attributed by a ranking that satisfies (A) and (B). If the discount factor  $\beta$  satisfies  $\beta \leq 1 - K$ , then the  $q$ -value estimates will converge weakly towards the smoothed best-response  $q$ -values  $Q^{*i}$ . Moreover, the global policy defined by  $\pi^{*i} = \phi(Q^{*i})$  for all  $i$  is an equilibrium.

Equation (3.3) means that for any learning rates attribution satisfying (A) and (B), convergence to a best-response equilibrium is preserved when every agent only receives a local reward gathered from its neighbors. The iterates (3.3) will be labeled **NetworkMQL**, and the proof is in the Supplement (Subsection 4.1).

Furthermore, convergence can also be ensured when the ranking only satisfies hypothesis (A) but not (B), provided that a slightly stronger contraction assumption is true. Let us consider such a ranking  $\tilde{r}k$  taking values in  $\{1, \dots, \tilde{M}\}$ .

A 3.3. Let  $Q' = (Q^{\mathcal{V} \setminus (i, \mathcal{N}^{\mathcal{D}(i)})}, Q^i, Z^{\mathcal{N}^{\mathcal{D}(i)}}(Q^i, Q^{\mathcal{V} \setminus (i, \mathcal{N}^{\mathcal{D}(i)})}))$  and  $|\tilde{r}k(i)|$  the number of agents with rank  $\tilde{r}k(i)$ . With the same notations as A 3.1, for every agent  $i$  in  $\mathcal{G}$  there exists  $K \in (0, 1)$  such that  $|r_\pi^i - r_{\pi'}^i| \leq \frac{K}{|\tilde{r}k(i)|} \|Q^i - Q^i\|_\infty$ .

Note that as for A 3.2, when  $\phi$  returns a Boltzmann distribution, A 3.3 is verified when the maximum change in  $q$ -values for all local best-responses in  $Q^{>i}$  is not greater than the change in  $Q^i$ , and the reward is scaled by a corresponding factor. Then, convergence towards a best-response equilibrium is preserved:

Theorem 3.3. Consider  $M$  agents locally updating their  $q$ -values estimates according to iterates (3.3) in a ND-POMDP, with learning rates  $\{\alpha_k\}$  and DAG interaction network A 3.3. The learning rates are attributed by a ranking that satisfies (A). If the discount factor  $\beta$  satisfies  $\beta \leq 1 - K$ , then the  $q$ -value estimates will converge weakly towards the smoothed best-response  $q$ -values  $Q^{*i}$ . Moreover, the global policy defined by  $\pi^{*i} = \phi(Q^{*i})$  for all  $i$  is an equilibrium.

The proof of this theorem is a straightforward application of Theorem 5.2. We can analyze the behavior of  $\tilde{M}$  iterates, where each iterate  $j \in 1, \dots, \tilde{M}$  is updating the  $q$ -tables of all agents of rank  $j$ . We provide a sketch of the proof in the Supplement (Subsection 4.2). Theorem 3.3 allows us to derive faster algorithms, as the number of ranks  $\tilde{M}$  needed to satisfy (A) is naturally lower than the number of ranks to satisfy both (A) and (B) ( $\tilde{M} \leq \bar{M}$ ): it suffices to give to every node the size of the longest path from any root to this node in the DAG.

**4. Application to wind farm control.** We now apply our multi-scale approach to an industrial problem: wind farm control. Wind turbines are often grouped together on the same field, in what are known as wind farms. Yet an operating wind turbine causes local wind perturbations - called wake effects - that can reduce the production of its neighbors. A solution is to misalign the rotor of a turbine with respect to the incoming wind: this can

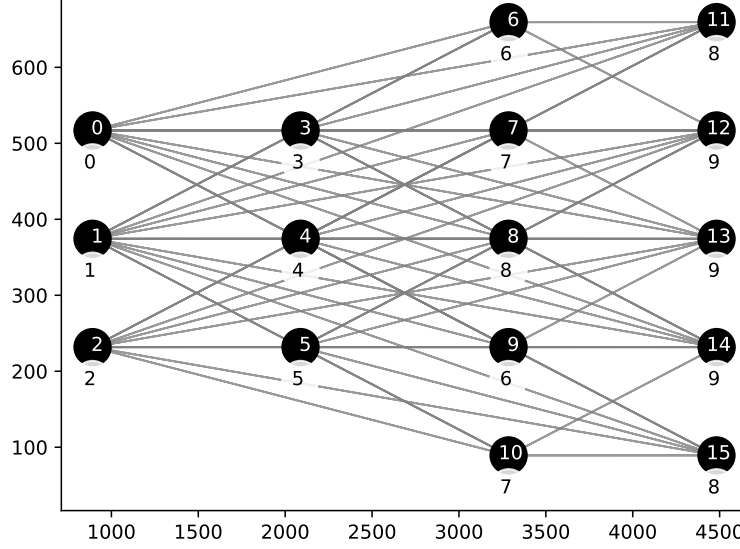


Figure 1: 16 interacting wind turbines on a graph. The levels used in the MQL algorithm are written in white, and the corresponding levels used in the NetworkMQL algorithm obtained with Algorithm 4.1 are written in black. The coordinates represent the location of each turbine in the farm.

diminish the impact of its perturbations on its neighbors, but at the cost of decreasing its own production. Finding a control strategy that can maximize output production can therefore be described as a multi-agent cooperative optimization problem, where every turbine is an agent. The application of multi-agent reinforcement learning strategies to this problem has been a growing topic of interest in the literature [2, 20, 9]. In this section, we show how the multi-scale Q-learning algorithm introduced in Section 3 can be used to solve this problem. The first step is to show the problem can be reformulated as a TI Dec-MDP by a careful design of the local observation spaces.

**4.1. Modeling a TI Dec-MDP.** We consider a farm of  $M$  wind turbines, where each turbine is controlled by an agent. The angle between a turbine's rotor and the direction of the wind is called the yaw, and we write  $y = (y^1, \dots, y^M)$  the vector of yaws of the  $M$  turbines. We set the local action spaces  $A_i$  to be the choice of increasing or decreasing a yaw by  $1^\circ$ , or to let it unchanged, so that  $A_i = \{-1, 0, +1\}$ . The finite global action space is then  $A = \prod_i A_i$ . The individual production of the  $i$ th turbine is given by  $\mathcal{P}(\mathbf{w}^i, y^i)$ , where  $\mathcal{P}$  is a function mapping local wind conditions  $\mathbf{w}^i$  and yaw  $y^i$  to a power output. The total production of the wind farm is then  $\mathcal{P}_{farm} = \sum_i^M \mathcal{P}(\mathbf{w}^i, y^i)$ . This problem can be framed as a MDP if the state  $s$  of the system contains both  $y$  and a sufficient statistics of the evolution of wind conditions in the farm, like the full wind field. The reward is then simply the total

production in a given state. Yet such a global state is not observed, as agents at best receive local observations of the wind field  $\mathbf{w}^1, \dots, \mathbf{w}^M$ . We show that the problem can rather be framed as a transition-independent Dec-MDP.

Let us assume that statistics on the wind inflow  $\mathbf{w}^\infty$  at the entrance of the farm can be represented by an irreducible and aperiodic Markovian process taking values in a finite state space with transition kernel  $P_W$ .  $\mathbf{w}^\infty$  is of course not controllable by the agents, it is an exogenous process. Moreover, all  $\mathbf{w}^i$  can be written as a function of  $\mathbf{w}^\infty$  and  $y$ , since they are the product of the perturbations caused by the turbines' wakes to the wind field. We can write this cooperative problem as a TI Dec-MDP, if we pick local states to be  $s^i = (y^i, \mathbf{w}^\infty) \in S_i$  and the finite global state space  $S = \Pi_i S_i$ . The reward  $r(s, a)$  returns the total production of the farm after agents have picked action  $a$  in state  $s$ . The transition function is  $P = P_y P_W = \Pi_{i=1}^M P_y^i P_W$ , where  $P_y^i$  is the transition kernel on the local yaw. Note that  $P_y^i$  is then entirely deterministic as for any  $s^i, a^i, s'^i \in S_i \times A_i \times S_i$  we have  $P(s^i, a^i, s'^i) = I\{s'^i = s^i + a^i\}$ . It is easy to see that if all local policies are forced to maintain non-null probabilities on all local actions, then the local state processes will be irreducible and aperiodic. Note that if we had allowed agents to observe their local wind conditions  $\mathbf{w}^i$ , the problem would not be transition-independent: any action taken by an agent can change the wind conditions at other agent's locations. This redesign of the local state is therefore a crucial step to analyze our problem. We now discuss how additional knowledge about interaction between turbines can guide both reward design and learning rates attribution.

**4.2. Learning rates attribution.** A DAG modeling interactions between agents can be built the following way: from  $M$  nodes representing the  $M$  agents, we add an edge from  $i \rightarrow j$  if turbine  $j$  is in the wake of turbine  $i$ . We then rewrite the reward as a sum of local components  $r(s, a) = \sum_i^M r^i(s^i, a^i, s^{U^i}, a^{U^i})$ , where each  $r^i$  returns the production of agent  $i$ , and  $U^i$  is the set of in-neighbors of turbine  $i$ . Then, A 3.2 essentially means that a small change to the stationary distribution of turbine  $i$ 's yaw will not lead to a greater change in average production, once the subset of  $i$ 's descendants have adjusted. This is a reasonable assumption as, first, small changes in turbine actuation do not lead to drastic changes in power output, and second, best-response adaptations require less policy changes as distance to  $i$  in the field increases. We use a topological sorting algorithm to attribute  $M$  different ranks to the agents (black circles on Figure 1), and our procedure (Algorithm 4.1 in Subsection 4.1) to attribute ranks satisfying (A) and (B) on this DAG (white circles in Figure 1).

We start by defining  $M$  learning sequences: for each rank in  $\{1, \dots, \bar{M}\}$ , let  $0 < l_{\bar{M}} < \dots < l_1 < 1$  and the corresponding learning rate sequences be

$$\alpha_{k,c}^{l_i} = \frac{g}{n_k((s^i, a^i)_c)^{l_i}}$$

with  $g > 0$  a gain and  $n_k((s^i, a^i)_c) = \#$  visits to the  $c$ th state-action pair  $(s^i, a^i)_c$  up to  $k$ . These sequences are standard for Q-learning algorithms. For our multi-scale experiments, we modify the sequences by adding a term dependent on the time between visits, so that the final learning rate sequences are:

$$\alpha_{k,c}^{l_i} = g \left( \frac{1}{n_k((s^i, a^i)_c)} + \frac{\log(T_{k,c}^i) - \log(T_{k-1,c}^i)}{T_{k,c}^i - T_{k-1,c}^i} \right)^{l_i}$$

where  $T_{k,c}^i$  is the real time of the  $k$ th update to component  $c$ . This second term is motivated by the theory of convergence of asynchronous iterates (see A 5.5 in Subsection 5.1): used alone, it ensures that learning happens at the same time-scale for all components of a single q-table. The first term used alone, or  $\alpha_{k,c}^{l_i}$ , ensures on the other hand that different agents learn at different time-scales. An example of the evolution of the resulting sequences of learning rates for Algorithm *MQL* can be found on Figure 2c, and we can see empirically that they preserve both properties. We use the same gain  $g = 2$  for all agents in all algorithms.

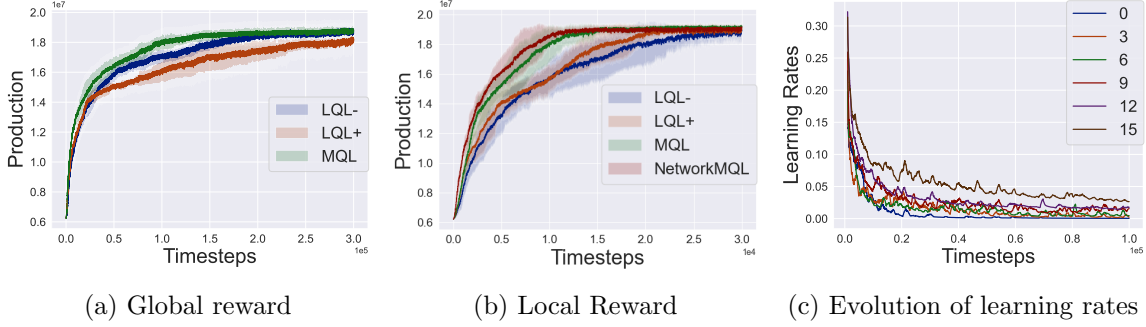


Figure 2: Total power output of the simulated 16 turbines wind farm averaged on 1000 time-steps under global (Figure 2a) and local (Figure 2b) reward. The evolution of learning rates under *MQL* based on scales attributed in Figure 1 is reported on Figure 2c for the first 100k time-steps.

**4.3. Results.** We run both Algorithm *MQL* (3.1) and Algorithm *NetworkMQL* (3.3) on a simulation of a wind farm with 16 wind turbines on 10 different seeds. We report the average production and standard deviation over the 10 seeds on Figure 2. Experiments are ran for both the global reward (Figure 2a), and the local reward (Figure 2b) derived from the interaction graph as described in Subsection 4.1. *NetworkMQL* is only run on the local reward, as local decomposition is a requirement for its convergence. In both cases, we compare these results with a naive local Q-learning approach denoted *LQL*, where a single-agent Q-learning algorithm is run at every agent with the standard learning rates sequences  $\alpha_{k,c}^{l_i}$ . For *MQL*, we simply assign a different rank to every agent following a topological sort and use the  $M$  multi-scale learning rate sequences  $\alpha_{k,c}^{l_i}$ , where the  $\{l_i\}_{i \in 1 \dots M}$  are taken at equal interval within  $[0.51, 1]$ . For *NetworkMQL*, we seek to assign  $\bar{M} \leq M$  ranks to all agents in the DAG satisfying hypotheses (A) and (B) in Subsection 3.2. We obtain  $\bar{M} = 9$  different ranks shown in Figure 1 and use the last 9 learning rate sequences in  $\{l_i\}_{i \in 1 \dots M}$ . We evaluate the performance of *LQL* with both  $l_i = 1$  (*LQL-*) and  $l_i = 0.51$  (*LQL+*), corresponding to the slowest and fastest learning rate sequence considered in all experiments.

In the global reward case (Figure 2a), *MQL* tracks or outperforms *LQL* on the different learning rate sequences. Note that finding the best performing learning rate sequence for *LQL* can require significant computing time, and that this approach provides no convergence guarantee. Then, exploiting local interaction (Figure 2b), *NetworkMQL* further accelerates

convergence on both the naive  $LQL$  and  $MQL$ .

**5. Proofs: Convergence of multi-scale Q-learning iterates.** To derive the convergence of the multi-scale Q-learning iterates, we exploit weak convergence results for stochastic approximation iterates with state-dependent noise [21]. Weak convergence results for synchronous two-timescale updates are laid out in [Theorem 6.2, Chapter 8, [21]], and for single-timescale asynchronous updates in [Theorem 5.1, Chapter 12, [21]]. We extend these results to asynchronous multi-timescale updates in Subsection 5.1. The formal proof of this extension has no technical difficulty but is tedious, and thus we detail it in the supplementary material: we first extend the weak convergence results for synchronous two-timescale to synchronous multi-timescale iterates, then use [Theorem 5.1, Chapter 12, [21]] to further extend this result to asynchronous multi-timescale updates (see Theorem 5.1).

We then prove in Subsection 5.2 that our multi-scale Q-learning converges under carefully chosen learning rates (Theorem 3.1). More precisely, we first show in Theorem 5.2 that our multi-scale Q-learning algorithm can be framed as multi-scale stochastic approximation updates with state-dependent noise, where the tracking error of the global state can be modeled as a latent Markovian process and satisfies necessary assumptions for applying Theorem 5.1. From Theorem 5.1 it follows that the multi-scale Q-learning updates can be approximated by  $M$  ODE: we give their expressions and analyze their convergence under Assumption A 3.1 in Subsection 5.3.

**5.1. Weak convergence of multi-scale iterates with Markovian noise.** We consider the constrained case: at each iteration, the iterates are projected on a defined admissible space  $H$ . We assume that  $H$  is a hyperrectangle  $H = [h_1, b_1] \times [h_2, b_2] \times \cdots \times [h_d, b_d]$  with  $(h_i, b_i) \in \mathbb{R}^2$  for  $i \in \{1, \dots, d\}$  and  $d > 0$  the dimension of the iterate. The operator  $\Pi_H$  is used to denote this projection on  $H$ .

Consider  $M$  interdependent stochastic approximation processes  $\theta_k^1, \dots, \theta_k^M$  updated according to iterates:

$$(5.1) \quad \theta_{k+1}^i = \Pi_H [\theta_k^i + \alpha_k^i Y_k^i] = \theta_k^i + \alpha_k^i (F^i(\theta_k, \xi_k^i) + \delta U_k^i) + B_k^i$$

where  $\theta_k = (\theta_k^1, \dots, \theta_k^M)$ ,  $\{\xi_k^i\}$  are noise sequences,  $F^i(\cdot, \cdot)$  are functions of  $\theta$  and  $\xi^i$ ,  $\delta U_{k+1}^i = Y_k^i - F^i(\theta_k, \xi_k^i)$  are martingale noise differences,  $\alpha_k^i := \alpha^i(k) > 0$  are learning rates for timescale  $i$  at iterate  $k$ , and  $B_k^i$  is a correction term to project the iterate on  $H$ , henceforth referred as reflection terms.

We assume that the iterates are updated asynchronously: that is, not all elements of the  $\theta^i$  are updated at every iteration. We index elements in every  $\theta^i$  by  $c \in \{1 \dots C\}$ . Let  $\alpha_{k,c}^i$  be the learning rate for element  $c$  of iterate  $i$  at timestep  $k$ : all elements within a single iterate are given the same sequences of learning rates, so that we use the notation  $\alpha_k^i = \alpha_{k,1}^i = \alpha_{k,2}^i = \cdots = \alpha_{k,C}^i$ . The  $M$  iterates in (5.1) can therefore be seen as  $M \times C$  iterates, with the updates to each component following:

$$(5.2) \quad \theta_{k+1,c}^i = \Pi_H [\theta_{k,c}^i + \alpha_{k,c}^i Y_{k,c}^i]$$

The time between the  $k$ th and  $(k+1)$ th updates of the element indexed by  $c$  in  $\{\theta_k^i\}_k$  is given by the random variable  $\tau_{k,c}^i$ . Because the  $k$ th update can happen at a different time for

two components, we need another timeline to analyze the behavior of the iterates. We will look at their behaviors in the "real time", so that the  $k$ th update at element  $c$  in the  $\{\theta_k^i\}_k$  is done at the real time  $T_{k,c}^i = \sum_{n=0}^{k-1} \tau_{n,c}^i$ .

Two sets of random variables therefore need to be considered at every iteration:  $Y_{k,c}^i$  and  $\tau_{k+1,c}^i$ . Following [21], we define two corresponding sequences of non-decreasing  $\sigma$ -algebras generated by the two sequences  $\mathcal{F}_{k,c}^{i,\tau}$  and  $\mathcal{F}_{k,c}^{i,Y}$ : each measures all variables observed in the "past" up to the relevant moment during update  $k+1$ . The associated conditional expectations are noted  $\mathbb{E}_{k,c}^{i,\tau}$  and  $\mathbb{E}_{k,c}^{i,Y}$ .

We now lay down the assumptions needed to ensure convergence. We start by standard assumptions for stochastic approximation algorithms: the sequences of observations  $Y_{k,c}^i$  are uniformly integrable, and at each timestep their expectations are given by a continuous function of the iterate  $\theta_{k,c}^i$ . The main idea is that an error term  $\delta U_{k,c}^i$  of null expectation will be averaged out through the iterations, so that as  $k$  goes to infinity, the behavior of the iterates can be described without the error terms. Let us write  $\xi_k^i = (\xi_{k,1}^i, \dots, \xi_{k,C}^i)$ , and  $\delta U_k^i = (\delta U_{k,1}^i, \dots, \delta U_{k,C}^i)$ . We make the following assumption for  $i = 1, \dots, M$ , with any statement on a sequence  $X_k^i$  interpreted as holding for all component-wise sequences  $X_{k,c}^i$ . Let  $\Xi$  be a complete and separable metric space, and  $A$  be an arbitrary compact set in  $\Xi$ .

**A 5.1.** *The  $\{Y_k^i\}$  are uniformly integrable, and can be written  $Y_k^i = F^i(\theta_k, \xi_k^i) + \delta U_k^i$  with  $\{\delta U_k^i\}_k$  martingale noise differences  $\mathbb{E}_k^Y \delta U_k^i = 0$  and  $F^i(\cdot, \xi^i)$  functions continuous in  $\theta$ , and continuous in  $\xi^i \in A$ .*

Here,  $F^i$  is still dependent on the error sequences  $\xi_k^i$  whose expectations are not null. Yet, the Markovian property of these sequences, combined with a constraint on the rate of evolution of the learning rates (see A 5.5), can be exploited to construct an approximation of  $F^i(\cdot, \xi^i)$  that does not depend on  $\xi^i$ . We make the following assumptions on the Markovian noise processes.

**A 5.2.** *The noise processes  $\{\xi_k^i\}$  are bounded with values in  $\Xi$ , and are Markovian: they admit a transition function  $P^i(\cdot, \cdot | \theta)$  such that  $P^i(\cdot, A | \theta)$  is measurable for each Borel set  $A \subset \Xi$ , and  $P^i(\xi_{k+1}^i \in \cdot | \mathcal{F}_k) = P^i(\xi_k^i, \cdot | \theta_k)$ . This transition function is continuous and does not depend on  $k$ . For any compact  $A \in \Xi$  and  $\mu \in (0, 1)$ , there exists a compact  $A'$  such that  $P(\xi_{k+1}^i \in A' | \xi_k^i) \geq 1 - \mu$  for all  $\xi_k^i \in A$ .*

We now define the fixed- $\theta$  chain  $\{\xi_k(\theta)\}$ , the Markov chain on state space  $\Xi$  with the fixed transition function  $P(\xi, \cdot, | \theta)$ . It is the noise process starting from  $k$  if  $\theta$  stayed constant, i.e.,  $\{\xi_{k+j}(\theta), j \geq 0, \xi_k(\theta) = \xi_k\}$ . The continuous function of the actual noise process can be approximated by the continuous function of the fixed- $\theta$  chain process  $F^i(\cdot, \xi_k^i)$  (see the proof of Theorem 8.4.3 [21], p.271-275) if the rate of change of the learning rates is slow enough. If we can construct a function  $\hat{F}^i(\cdot)$  of  $\theta$  that does not depend on the process  $\xi^i$ , such that  $\hat{F}^i(\theta)$  is a local average of the  $F^i(\cdot, \xi_k^i)$ . Then  $\hat{F}^i(\theta_k)$  is also an approximation of  $F^i(\theta_k, \xi^i)$  as  $k \rightarrow \infty$ . We detail these assumptions in ?? 5.3?? 5.4.

**A 5.3.** *The set  $\{F^i(\theta_k, \xi_k^i)\}_k$  is uniformly integrable. For any  $j \geq 0$  with  $\xi_j^i \in A$  the set  $\{F^i(\theta, \xi_{j+k}^i(\theta))\}_{k \geq 0}$  is uniformly integrable, where  $\xi_{j+k}^i(\theta)$  is the fixed- $\theta$  chain with initial conditions  $\xi_j^i$  and transition function  $P^i(\xi, \cdot | \theta_j)$ .*

585 A 5.4 (Averaging condition). *There exists a continuous function  $\bar{F}^i(\cdot)$  such that for each*  
 586  *$\theta$  and on any compact set  $A \in \Xi$*

$$587 \quad \lim_{(k,m) \rightarrow \infty} \frac{1}{m} \sum_{j=k}^{k+m-1} \mathbb{E}_k^Y [F^i(\theta, \xi^i(\theta)) - \bar{F}^i(\theta)] I_{\xi_k^i \in A} = 0$$

588 We establish now a different timescale to correspond to each process. For  $i, j \in \{1, \dots, M\}$ ,  
 589 let  $t_k^j = \sum_{l=0}^{k-1} \alpha_l^j$ , and  $\theta_{c,\alpha^j}^{i,0}(t)$  be the piecewise interpolation of the process  $\theta_k^i$  on the  $j$ -th  
 590 timescale defined as

$$591 \quad \theta_{c,\alpha^j}^{i,0}(t) = \theta_{0,c}^i, \quad t \leq 0, \quad \theta_{c,\alpha^j}^{i,0}(t) = \theta_{k,c}^i, \quad t \in [t_k^j, t_{k+1}^j]$$

592 Then, a shifted continuous time interpolation  $\theta_{\alpha^j}^{i,k}(\cdot)$  is simply an interpolation "started" from  
 593 a specific time-step  $k$ . Since all components do not reach a given timestep at the same time,  
 594 we rather define the shifted interpolates as starting at arbitrary real times  $v$ . For this purpose,  
 595 we introduce functions  $p_c^i(v)$ , that return the time-step of the first update of  $\theta_c^i$  after a given  
 596 real time  $v$ . The shifted interpolates at the  $j$ -th timescale are then:

$$597 \quad (5.3) \quad \theta_{c,\alpha^j}^{i,v}(t) = \theta_{k+p_c^i(v),c}^i \quad t \in [t_{k,c}^{ij,v}, t_{k+1,c}^{ij,v}), \quad t_{k,c}^{ij,v} = \sum_{n=p_c^i(v)}^{k-1} \alpha_n^j$$

598 and the shifted real-time interpolations  $\hat{\theta}_{c,\alpha^j}^{i,v}(\cdot)$  are defined similarly:

$$599 \quad (5.4) \quad \hat{\theta}_{c,\alpha^j}^{i,v}(t) = \theta_{k,c}^i \quad t \in [\Gamma_{k+p_c^i(v)}^{ij,v}, \Gamma_{k+1,c}^{ij,v}) \quad \Gamma_{k,c}^{ij,v} = \sum_{n=p_c^i(v)}^{k-1} \alpha_n^j \tau_{n,c}^i$$

600 Similarly, we define  $B_{c,\alpha^j}^{i,v}$  (resp.  $\hat{B}_{\alpha^j}^{i,v}$ ) the shifted continuous (resp. real time) interpolations  
 601 at the  $j$ -th timescale of the sequence of reflection terms  $B_{c,k}^i$ . We are interested in the behavior  
 602 of  $\theta_{\alpha^j}^{i,v}(\cdot)$  and  $B_{\alpha^j}^{i,v}(\cdot)$  as  $t_k^j \rightarrow \infty$  while  $\alpha_k^j \rightarrow 0$ .

603 We now lay out in A 5.5 further constraints on the learning rate sequences. The first two  
 604 are standard for the stochastic approximating literature: intuitively they require the learning  
 605 rates to go towards zero, but not too quickly. The third assumption ensures that even though  
 606 the components of a single iterate are updated asynchronously, they still learn at the same  
 607 timescale. Finally, the last assumption makes the iterates multi-scale: it imposes a hierarchy  
 608 between the  $M$  sequences that ensures every iterate is learning at a different timescale.

609 A 5.5 (Assumption on learning rates). *For each  $i \in \{1, \dots, M\}$ ,*

- 610 • (Classical rates)  $\lim_k \alpha_k^i = 0$  and  $\sum_{k=0}^{\infty} \alpha_k^i = \infty$  with  $\alpha_k^i \in (0, 1)$  for all  $i, k$
- 611 • (Slow changes) there is a sequence of integers  $a_n^i \rightarrow \infty$  such that

$$612 \quad \lim_n \sup_{0 \leq j \leq a_n^i} \left| \frac{\alpha_{n+j}^i}{\alpha_n^i} - 1 \right| = 0$$

- (Asynchronous) Every component's learning rate  $\alpha_{k,c}^i$  can be written as a local average of positive real-valued functions  $f^i$ :

$$\alpha_{k,c}^i = \frac{1}{\tau_{k,c}} \int_{T_{k,c}^i}^{T_{k,c}^i + \tau_{k,c}^i} f^i(s) ds \quad \text{such that} \quad \int_0^\infty f^i(s) ds = \infty \quad \text{and} \quad \lim_{s \rightarrow \infty} f^i(s) = 0$$

- (Multi-scale)  $\frac{\alpha_k^i}{\alpha_k^j} \rightarrow 0$ , as  $k \rightarrow \infty$ , whenever each  $i < j$ .

Finally, all components of an iterate should be updated frequently enough. This is formalized below:

A 5.6. For all  $i$ , the sequence of intervals between updates  $\{\tau_{k,c}^i\}_k$  is uniformly integrable, and there exists  $\bar{u}_c^i \geq 1$  such that the  $\mathbb{E}_{k+1,c}^{i,\tau} [\tau_{k,c}^i]$  are in the bounded interval  $[1, \bar{u}_c^i]$  uniformly in  $k$ .

With the expectations  $\mathbb{E}_k^Y Y_k^i$  being approximated by  $\bar{F}^i(\theta_k)$  as  $k$  goes to  $\infty$ , the interpolations of the iterates  $\theta_{\alpha^j}^{i,v}$  will be shown to admit limit processes following mean ODEs defined by the  $\bar{F}^i$ . The solution of the ODE can then be used to characterize the asymptotic properties of the  $\theta_{k,c}^i$  for  $i = 1, \dots, M$ . Thanks to the multi-scale assumption, at any timescale  $j$  the interpolation for all iterates learning at a slower timescale  $i < j$  will follow the null ODE. Intuitively, they evolve so slowly that they can be considered constant at the  $j$ -th timescale. Similarly, the interpolations for all iterates learning at a faster timescale can be considered to have reached the limit of their respective mean ODE, if it exists. We consider the case where the ODE for every limit process for any timescale has a unique asymptotically stable point. Adopting like in Subsection 3.1 the notations  $\theta^{>i} = (\theta^{i+1}, \dots, \theta^M)$  and  $\theta^{<i} = (\theta^1, \dots, \theta^{i-1})$ :

A 5.7. There exists a continuous function  $\zeta^i(\theta^{<i})$  such that, for any set of initial conditions  $\theta$ , the solution to the following ODE has a unique asymptotically stable point  $(\theta^{<i}, \zeta^i(\theta^{<i}))$  for  $i \geq 2$ :

$$\begin{aligned} \dot{X}^j &= 0 \quad \text{for } j < i \\ \dot{X}^i &= \frac{\bar{F}^i(X^{<i+1}, Z^{\geq i+1}(X^{<i+1}))}{u_c^i} + \hat{b}^i. \end{aligned}$$

with  $u_c^i(t)$  taking values in  $[1, \bar{u}_c^i]$ ,  $\hat{b}^i$  the reflection term on  $H$ , and

$$(5.5) \quad Z^{\geq i}(\theta_k^{<i}) = (\zeta^i(\theta_k^{<i}), Z^{\geq i+1}(\theta_k^{<i}, \zeta^i(\theta_k^{<i}))), \quad i = 2, \dots, M-2$$

with  $Z^{\geq M-1}(\theta_k^{<M-1}) = (\zeta^{M-1}(\theta_k^{<M-1}), \zeta^M(\theta_k^{<M-1}, \zeta^{M-1}(\theta_k^{<M-1})))$ .

When applying multi-scale iterates (5.2) to our Dec-POMDP problem, this assumption imposes strong constraints on the dynamics of the multi-agent system. In Section 3, we introduced specific DAG structures on agent interactions that can satisfy them, and a concrete example was given in Section 4.

Note that the reflection terms  $b^i$  of the projected ODE must live within a convex space  $\Upsilon(X^i)$ , defined the following way: on the interior of  $H$ ,  $\Upsilon(X^i) = \{0\}$ , the set only containing

the null vector, and on the boundary of  $H$ ,  $\Upsilon(X^i)$  is the infinite convex cone generated by the outer normals at  $X^i$  of the faces on  $H$  on which  $X^i$  lies.

Now we state the weak convergence of the iterates (5.2) in the following theorem.

**Theorem 5.1** (Weak convergence of asynchronous multi-scale iterates with Markovian noise). *Consider iterates (5.2), updated asynchronously following the time interval sequences  $\{\tau_{k,c}^i\}_k$ . If Assumptions ?? 5.1–5.7 hold, then for a sequence of real times  $\{\nu_k\}_k$  such that  $\nu_k \rightarrow \infty$  as  $k \rightarrow \infty$ ,  $\hat{\theta}_{c,\alpha^1}^{1,\nu_k}$  admits a subsequence which converges towards a process  $\hat{\theta}_{c,\alpha^1}^1(\cdot)$  such that:*

$$\dot{\hat{\theta}}_{c,\alpha^1}^1(t) = \frac{\bar{F}^1(\theta_{c,\alpha^1}^1(t), Z^{\geq 2}(\theta_{c,\alpha^1}^1(t)))}{u_c^i(t)} + \hat{b}_{c,\alpha^1}^1(t) \quad u_c^i(t) \in [1, \bar{u}_c^i] \quad \hat{b}_{c,\alpha^1}^1(t) \in -\Upsilon(\hat{\theta}_{c,\alpha^1}^1(t)) \quad (5.6)$$

where  $\hat{b}_{c,\alpha^1}^1$  is the reflection, that is the minimum force needed to keep  $\hat{\theta}_{c,\alpha^1}^1$  in  $H$ . Moreover, for any  $\delta > 0$ , the fraction of time spent by  $\hat{\theta}_{c,\alpha^1}^1(\cdot)$  in any  $\delta$ -neighborhood around the set of limit points of (5.6) on the interval  $[0, T]$  goes to one in probability as  $T \rightarrow \infty$ .

With the weak convergence of the multi-scale iterates established, we are now ready to prove the weak convergence of our multi-scale Q-learning iterates (3.1).

## 5.2. Application to multi-scale Q-learning iterates.

**Theorem 5.2.** *Consider the multi-scale Q-learning iterates (3.1). If ?? 5.5?? 5.7 are true, let all  $\hat{Q}_{0,c}^i \in [-D, D]$  for  $D > 0$  such that  $D > \frac{R}{1-\beta}$  with  $R$  the reward bound. The iterates  $\hat{Q}_{k,c}^i$  admit subsequences that converge towards the following mean ODEs as  $k$  goes to infinity:*

$$\frac{d}{dt} q_t^i(s_c^i, a_c^i) = \frac{1}{u_c^i} \bar{F}^i(q_t^{<i}, q_t^i, Z^{\geq i+1}(q_t^{<i+1}))$$

where

- the  $\{q_t^j\}_{j < i}$  are constant
- $\{q_t^j\}_{j > i} = Z^{>i+1}(q_t^{<i+1})$
- $\pi^{-i} = (\phi(q_t^1), \dots, \phi(q_t^{i-1}), \phi(q_t^{i+1}), \dots, \phi(q_t^M))$
- $\bar{F}_c^i(Q) = \sum_s d^{\pi_k}(s | s^i) \sum_{a^{-i}} \pi_k^{-i}(a^{-i} | s) r(s, a^i, a^{-i}) + \beta v^i(Q^i, (s^i, a^i)_c) - Q_c^i F_c^i(Q, \xi)$

The full proof of the theorem is given in the Supplement (Section 2). We first rewrite iterates (3.1) as the multi-scale stochastic approximation iterates (5.2). To achieve this we write the terms in brackets  $r_k + \beta[\phi(\hat{Q}_k)(\cdot | s_{k+1}^i)]^T \hat{Q}_k(s_{k+1}^i, \cdot) - \hat{Q}_k^i(s_k^i, a_k^i) := Y_{k,c}^i$  as the sum of  $\bar{F}_c^i(\hat{Q}_k)$  and two errors: (1)  $Y_{k,c}^i - \mathbb{E}_{k,c}^Y[Y_{k,c}^i]$  and (2)  $\mathbb{E}_{k,c}^Y[Y_{k,c}^i] - \bar{F}_c^i(\hat{Q}_k)$ . After noticing that (1) is a martingale sequence, the core of the proof is the analysis of the evolution of (2) for a fixed- $Q$  chain. It is shown that its local averages converge in mean to zero, satisfying A 5.4. This allows us to apply the results of Theorem 5.1 to the multi-scale Q-learning iterates (3.1).

**5.3. Proof of Theorem 3.1: Convergence with acyclic dependence structure.** Theorem 5.2 shows that under proper initialization of our Q-table estimates, the iterates of independent, multi-scale Q-learners can be approximated by an ODE defined by the  $M$  functions  $\bar{F}^i$ . To prove Theorem 3.1, it remains to analyze the behavior of these ODE and show that

A 5.7 is satisfied under Assumption A 3.1 on the acyclic dependence structure. Indeed, A 3.1 will guarantee the convergence of the approximating ODEs towards the desired equilibrium.

*Proof.* We recall the mean ODE followed by each agent  $i$  as introduced in (2.3):

$$\frac{d}{dt}q_t^i(s_c^i, a_c^i) = \frac{1}{u_c^i} \bar{F}^i(q_t^{<i}, q_t^i, Z_t^{\geq i+1}(q_t^{<i+1}))$$

with  $\bar{F}^i(q_t^{<i}, q_t^i, Z_t^{\geq i+1}(q_t^{<i+1})) = r(s_c^i, a_c^i, q_t^{<i}, q_t^i, Z_t^{\geq i+1}(q_t^{<i+1})) + \beta \sum_{s'} P(s^i, a^i, s'^i) q^i(s'^i, \phi(q)(s'^i))$  and  $r_{q^i}(s_c^i, a_c^i, q_t^{<i}, q_t^i, Z_t^{\geq i+1}(q_t^{<i+1})) = \sum_s d^{\pi_{q^i, Z^{\geq i+1}(q^i)}}(s) \sum_{a^{-i}} \pi_{q^i, Z^{\geq i+1}(q^i)}^{-i}(a^{-i}) r(s, a^i, s^{-i})$ . ■

According to A 3.1 and for each agent  $i$  and component  $c$ , the mapping from  $q^i$  to  $r_{q^i}(s_c^i, a_c^i, q^i)$  is a  $K$  - contraction mapping.  $\bar{F}^i$  is therefore a  $(K + \beta)$  contraction mapping with regard to the infinite norm. It follows that for each agent  $i$  there is a unique fixed point  $Q^{*i}$  such that  $\bar{F}^i(Q^{<i}, Q^{*i}, Z^{\geq i+1}(Q^{<i+1})) = Q^{*i}$  and that this fixed point is the unique globally asymptotically stable point of the ODE  $\dot{X} = \bar{F}^i(Q^{<i+1}, Z^{\geq i+1}(Q^{<i+1}))$ . Recall that the reflection terms are null. The multiplication by the factor  $1/u_c^i$  has a time scaling effect on the ODE but does not change its asymptotic behavior. It follows that A 5.7 on the asymptotic behaviors of the mean ODEs is satisfied. A sequence of learning rates  $\alpha_k^i$  has been assigned to each agent  $i$  such that A 5.5 are satisfied. The weak convergence of the iterates  $\hat{Q}_k$  towards a smoothed equilibrium then follows from Theorem 5.2. ■

**6. Conclusions.** By allowing all agents to run a single-agent reinforcement algorithm in parallel, independent learning provides the simplest way to adapt these algorithms to cooperative multi agent environments. Although this approach has encountered experimental successes, it has no underlying theoretical guarantee. To provide a first step towards bridging this gap, we have here focused on transition-independent Dec-PODMP, and shown that in these problems the partial observability of the global state can be modeled as a Markovian perturbation in a stochastic approximation iterates. We have shown that when there is an acyclic dependence structure between agent dynamics in these cooperative systems, a careful assignment of learning rate sequences following a multi-scale approach can be sufficient to establish convergence. In particular, knowledge of the interaction graph between agents in ND-POMDP can be exploited to assign learning rates to preserve convergence. We have then applied these results to wind farm control, a real optimization problem from the industry.

Further work can extend these results to systems with noisy local observations or non-independent transition functions. Furthermore, independent learning has often encountered experimental success without the multiscale approach in multiagent reinforcement learning settings ([42, 39]), and our acyclic dependence analysis could provide a basis to find a theoretical explanation of these results.

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## Supplementary Material.

**1. Proof of Recursive form of local q-function.** The expectation of the reward  $r(s, a)$  is taken with regard to a distribution over the global state and the global action. Because the q-value is evaluating the response  $\pi^i$  to  $\pi^{-i}$ , the global action must always be taken with respect to  $\pi$ . Then, per definition of the q-value (2.1), the initial state is sampled from the stationary distribution  $d^\pi$ . It then follows from the definition of the stationary distribution that the distribution of the next global state will still be  $d^\pi$ , and the local q-value taken at the next step is the expectation of the future gain. This is formalized below.

*Proof of Lemma 2.1 .* We can rewrite (2.1) as

$$(1.1) \quad Q_{\pi^i}^{\pi^{-i}}(s^i, a^i)$$

$$(1.2) \quad = \sum_s d^\pi(s \mid s^i) \mathbb{E}_{a_0 \sim (\pi^i, \pi^{-i})} [r(s_0, a_0) \mid s_0^i = s^i, a_0^i = a^i, s_0 = s]$$

$$(1.3) \quad + \beta \sum_s d^\pi(s \mid s^i) \mathbb{E}_{a_k \sim (\pi^i, \pi^{-i}), s_k \sim P} \left[ \sum_{k=1}^{\infty} \beta^{k-1} r(s_k, a_k) \mid s_0^i = s^i, a_0^i = a^i, s_0 = s \right]$$

While the term in (1.2) can be expressed as  $\sum_s d^\pi(s \mid s^i) \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid s) r(s, a^i, a^{-i})$ , the term in (1.3) can be developed in the following way

$$\begin{aligned} & \beta \sum_s d^\pi(s \mid s^i) \mathbb{E}_{s_1, a_1^i} \left[ \mathbb{E}_{a_k \sim \pi, s_k \sim P} \left[ \sum_{k=1}^{\infty} \beta^{k-1} r(s_k, a_k) \mid s_0 = s, a_0^i = a^i, s_1 = s', a_1^i = a'^i \right] \mid s_0 = s, a_0^i = a^i \right] \\ &= \beta \sum_s d^\pi(s \mid s^i) \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid s) \sum_{s'} P(s, a, s') \sum_{a'^i} \pi^i(a'^i \mid s'(i)) \times \\ & \quad \mathbb{E}_{a_k \sim \pi, s_k \sim P} \left[ \sum_{k=1}^{\infty} \beta^{k-1} r(s_k, a_k) \mid s_1 = s', a_1^i = a'^i \right] \\ &= \beta \sum_{s'^i} P^i(s^i, a^i, s'^i) \sum_{a'^i} \pi^i(a'^i \mid s'(i)) \left[ \sum_s d^\pi(s \mid s^i) \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid s) \sum_{s'} P^{-i}(s^{-i}, a^{-i}, s'^{-i}) \times \right. \\ & \quad \left. \mathbb{E}_{a_k \sim \pi, s_k \sim P} \left[ \sum_{k=1}^{\infty} \beta^{k-1} r(s_k, a_k) \mid s_1^i = s'^i, s_1^{-i} = s'^{-i}, a_1^i = a'^i \right] \right] \\ &= \beta \sum_{s'^i} P^i(s^i, a^i, s'^i) \sum_{a'^i} \pi^i(a'^i \mid s'(i)) \left[ \mathbb{E}_{s_1 \sim d^\pi, a_k \sim \pi, s_k \sim P} \left[ \sum_{k=1}^{\infty} \beta^{k-1} r(s_k, a_k) \mid s_1^i = s'^i, a_1^i = a'^i \right] \right] \\ &= \beta \sum_{s'^i} P^i(s^i, a^i, s'^i) \sum_{a'^i} \pi^i(a'^i \mid s'(i)) Q_{\pi^i}^{\pi^{-i}}(s'^i, a'^i) \end{aligned}$$

where the second line to the last is due to the definition of the stationary distribution.

**2. Proof of the weak convergence of multi-scale Q-learning (Theorem 5.2).** We present below the detailed proof of Theorem 5.2, in which we apply the results of Theorem 5.1 on the weak convergence of asynchronous multi-scale stochastic approximation iterates to our Q-learning iterates (3.1). The first step is to rewrite (3.1) as the iterates of Theorem 5.1, and

829 identify the corresponding martingale and Markovian noise. We then analyze the two error  
830 terms separately.

831 *Proof.* Let us consider any local state action pair  $s^i, a^i$  of any iterate  $Q^i$ . By assumption on  
832 the transition kernel A 2.2 and the design of the mapping  $\phi$ , the sequence of times between two  
833 visits are uniformly integrable. All return times must moreover be at least 1. The  $\mathbb{E}_{n+1,c}\tau_{n,c}^i$   
834 are therefore uniformly bounded with values in an interval  $[1, \bar{u}_c^i]$  with  $\bar{u}_c^i \geq 1$ , therefore  
835 satisfying A 5.6.

836 In the following, we write  $\{\bar{F}_c^i(Q)\}_c$  real-valued continuous functions for  $c \in \{1 \dots, |S_i||A_i|\}$ .  
837 and recall that

$$838 \quad (2.1) \quad \pi_k = (\pi_k^1, \dots, \pi_k^M) \quad \pi_k^j = \phi(\hat{Q}_k^j) \quad \hat{Q}_k = (\hat{Q}_k^1, \dots, \hat{Q}_k^M) \quad \hat{Q}_k^i = \{\hat{Q}_{k,c}^i\}_{c=1, \dots, C}$$

839 We can rewrite the iterates (3.1) in real time in the following way:

$$\begin{aligned} \hat{Q}_{k+1,c}^i &= \hat{Q}_{k,c}^i + \alpha_{k,c}^i I_{k,c} \left[ \bar{F}_c^i(\hat{Q}_k) + \delta U_{k,c}^i + \xi_{k,c}^i \right] \\ \delta U_{k,c}^i &:= Y_{k,c}^i - \mathbb{E}_{k,c}^Y[Y_{k,c}^i] \\ 840 \quad (2.2) \quad \xi_{k,c}^i &:= \mathbb{E}_{k,c}^Y[Y_{k,c}^i] - \bar{F}_c^i(\hat{Q}_k) \\ Y_{k,c}^i &:= r_k + \beta[\phi(\hat{Q}_k^i)(\cdot | s_{k+1}^i)]^T \hat{Q}_k^i(s_{k+1}^i, \cdot) - \hat{Q}_{k,c}^i \\ &= F_c^i(\hat{Q}_k, \xi_{k,c}^i) + \delta U_{k,c}^i \end{aligned}$$

841 where

$$\begin{aligned} \bar{F}_c^i(Q) &= \sum_s d^{\pi_k}(s | s^i) \sum_{a^{-i}} \pi_k^{-i}(a^{-i} | s) r(s, a^i, a^{-i}) + \beta v^i(Q^i, (s^i, a^i)_c) - Q_c^i \\ 842 \quad F_c^i(Q, \xi) &= \bar{F}_c^i(Q) + \xi \end{aligned}$$

with  $v^i(Q, (s^i, a^i)_c) = v^i(Q, s^i, a^i)$  for  $s^i, a^i$  the  $c$ th component of  $Q^i$  and recall that

$$v^i(Q^i, s^i, a^i) = \sum_{s'^i} P^i(s^i, a^i, s'^i) [\phi(Q^i)(\cdot | s'^i)]^T Q^i(s'^i, \cdot)$$

843 We first show that the iterates (2.2) are in fact equivalent to their constrained version:

$$844 \quad \hat{Q}_{k+1,c}^i = \Pi_{[-D,D]} \left( \hat{Q}_{k,c}^i + \alpha_{k,c}^i I_{k,c} \left[ \bar{F}_c^i(Q_{k,c}^i) + \delta U_{k,c}^i + \xi_{k,c}^i \right] \right)$$

845 Recall that at any update of the local state-action pair corresponding to the  $c$ th compo-  
846 nent, we have  $\hat{Q}_{k+1,c}^i = (1 - \alpha_{k,c}^i) \hat{Q}_{k,c}^i + \alpha_{k,c}^i \left[ r_k + \beta[\phi(\hat{Q}_k^i)(s_{k+1}^i)]^T \hat{Q}_k^i(s_{k+1}^i, \cdot) \right]$  for any  $i, k, c$ ,  
847 with  $\alpha_{k,c}^i \in (0, 1)$ . Moreover, here  $\phi(\hat{Q}_k^i)(s_{k+1}^i)$  defines a probability distribution and we have  
848 discount factor  $\beta \in (0, 1)$ . It follows that since  $\hat{Q}_{0,c}^i \in [-D, D]$  for all  $c$  and  $R + \beta D < D$ ,  
849 then we have  $\sup_k \|\hat{Q}_{k,c}^i\| < D$  for all  $c, k$  and the iterates will never leave the hyper-rectangle  
850 defined by  $[-D, D]^{|S_i||A_i|}$ . This means that for constrained iterates with constraint space  
851  $[-D, D]^{|S_i||A_i|}$ , the induced reflection term will always equal zero.

We have therefore shown that the iterates (3.1) can be written as the multi-scale stochastic approximation iterates (5.2). The  $Y_{k,c}^i$  are then uniformly bounded, and the  $F_c^i$  are moreover continuous in  $\xi$  and  $Q$ . Per definition, for any  $k, i, c$ ,  $\mathbb{E}_{k,c}^Y[\delta U_{k,c}^i] = 0$  and  $\{\sum_{j=0}^k \delta U_{k,c}^i\}_k$  is a martingale sequence, so that A 5.1 is satisfied.

If the noise sequences  $(\xi_{k,c}^i)$  satisfy ?? 5.2–5.4, then according to Theorem 5.1 the iterates follow the  $M \times C$  mean ODEs:

$$(2.3) \quad \frac{d}{dt} q_t^i(s_c^i, a_c^i) = \frac{1}{u_c^i} \bar{F}^i(q_t^{<i}, q_t^i, Z^{>i+1}(q_t^{<i+1}))$$

where  $\pi^{-i} = (\phi(q_t^1), \dots, \phi(q_t^{i-1}), \phi(q_t^{i+1}), \dots, \phi(q_t^M))$ , the  $\{q_t^j\}_{j < i}$  are constant,  $\{q_t^j\}_{j > i} = Z^{>i+1}(q_t^{<i+1})$ . Then, assumption A 5.7 guarantees that (2.3) admits an asymptotically stable point, and we conclude on the convergence of the iterates towards a smooth equilibrium as defined in Definition 2.3.

We now need to prove that the noise sequences  $\xi_{k,c}^i$  with values in the space  $\Xi$  defined in (2.2) are Markovian state-dependent noise sequences, satisfying ?? 5.2–5.4. We have  $F_c^i(\hat{Q}_k, \xi_k^i) = \mathbb{E}_{k,c}^Y[Y_{k,c}^i]$ . Since the  $Y_{k,c}^i$  are uniformly bounded, then so are the  $F_c^i(\hat{Q}_k, \xi_k^i)$ . As the arguments for the boundedness of the  $Y_{k,c}^i$  rely on the proper initialization of the  $\hat{Q}_0^i$  and the boundedness of the reward, they naturally transfer to any fixed- $Q$  chain sequence  $\{F_c^i(\hat{Q}_j, \xi_{j+k}^i(\hat{Q}_j))\}_k$ , so that A 5.3 is verified. Let us now derive an expression for  $\mathbb{E}_k^Y[Y_{k,c}^i]$ . First,  $\hat{Q}_{k,c}^i$  is a function of  $\hat{Q}_{0,c}^i$  and the previous  $Y_j^i, \tau_j^i, j < k$ , so we only need to focus on  $s_{k+1}^i$  and  $r_k$ . The next state  $s_{k+1}^i$  is sampled from the local transition kernel after the agent has visited component  $(s^i, a^i)_c$ , so we have exactly:

$$\mathbb{E}_{k,c}^Y \left[ [\phi(\hat{Q}_k^i)(s_{k+1}^i)]^T \hat{Q}_k(s_{k+1}^i, \cdot) \right] = v'(\hat{Q}_k^i, (s^i, a^i)_c)$$

As for the reward  $r_k$ :  $\mathbb{E}_{k,c}^Y r_k = \mathbb{E}_{k,c}^Y r(s_k, a_k) = \mathbb{E}_{k,c}^Y r((s^i, a^i)_c, s_k^{-i}, a_k^{-i})$ . Neither  $s_k^{-i}$  nor  $a_k^{-i}$  are observed by agent  $i$ . If  $s_k^{-i}$  was known however, then the expectation of  $a_k^{-i}$  would just be taken from the respective policies of other agents at that time  $\bar{\pi}^{-i} = \phi(\hat{Q}_k^{-i})$ :

$$\mathbb{E}_{k,c}^Y r_k = \mathbb{E}_{k,c}^Y \left[ \mathbb{E}_{k,c}^Y \left[ r((s^i, a^i)_c, s_k^{-i}, a_k^{-i}) \mid s_k^{-i} \right] \right] = \mathbb{E}_{k,c}^Y \left[ [\phi(\hat{Q}_k^{-i})(\cdot, s_k^{-i})]^T r((s^i, a^i)_c, s_k^{-i}, \cdot) \right]$$

It remains to handle  $s_k^{-i}$ . It is easy to see that the state process  $\{s_k^{-i}\}_k$  is in fact a Markovian process, whose transition kernel depends on the iterates  $\hat{Q}_k$  in real time. Recall that  $P$  is the global transition matrix of dimension  $|S| \times |A_i| \dots |A_M| \times |S|$ . By construction the mapping  $\phi$  returns a policy assigning a non-zero probability to every action, so that there exists  $\epsilon_\phi > 0$  such that for all  $a \in A_i$ ,  $\pi(a|s^i) > \epsilon_\phi$ . For an initial distribution  $d_0$ , we write  $\{d_k\}_{k \geq 0} \in (0, 1)^{|S|}$  the process tracking the distribution of  $s^{-i}$ :

$$d_{k+1} = d_k \cdot P \prod_{\substack{j=1 \\ j \neq i}}^M \phi(\hat{Q}_k^j)$$

$\{d_k\}_k$  is a state-dependent Markovian process, that is:

$$(2.4) \quad P(d_{k+1} \in \cdot | \mathcal{F}_{k,c}^{i,Y}, d_k) = P(d_{k+1} \in \cdot | \hat{Q}_k, d_k)$$

865 With a small abuse of notation we write  $d^{\pi_k}(s^{-i})$  for  $d^{\pi_k^{-i}}(s^{-i})$ , so that we have  $\{\xi_{k,c}^i\}_k$   
 866 as:

$$867 \quad (2.5) \quad \xi_{k,c}^i = \sum_s [d_k(s^{-i}) - d^{\pi_k}(s^{-i})] [\phi(Q_k^{-i})(\cdot, s_k^{-i})]^T r(s_c^i, s_k^{-i}, a_c^i, \cdot)]$$

868 where  $d^{\pi_k} = d^{\phi(Q_k)}$  is still the stationary distribution over global states under policy  $\phi(Q_k)$ .  
 869 Recall that under any fixed policies the local state processes follow independent dynamics, so  
 870 that the stationary distribution is factorized  $d^{\pi} = d^{\pi^i} d^{\pi^{-i}}$ .

871 Since the reward is bounded in  $[-R, R]$ , the  $\{\xi_{k,c}^i\}_k$  take values in the compact  $[-R|S||A|, R|S||A|]$ .  
 872 The  $\{\xi_{k,c}^i\}_k$  being an affine transformation of  $\{d_k\}$ , it follows that it is also a state-dependent  
 873 Markovian process. Moreover, this state-dependent process is stationary, in the sense that  
 874 for each  $Q$  there is a time-invariant (does not depend on  $k$  if we know  $Q$ ) measurable transi-  
 875 tion function  $P^\xi(\cdot, \cdot | Q)$  such that  $P(\xi_{k+1,c} \in \cdot | \mathcal{F}_{k,c}^{i,Y}, d_k) = P^\xi(\xi_{k,c}, \cdot | \hat{Q}_k)$ . Therefore, A 5.2 is  
 876 satisfied.

877 It remains to show that the noise  $\{\xi_i\}_k$  satisfies A 5.4: its "rate of change" is small enough  
 878 that it can be locally averaged out, and the noisy observations can be approximated by the  
 879 mean ODE. In particular, we define the fixed- $Q$  chain  $\{\xi_{k,c}(Q)\}$ , the Markov chain on state  
 880 space  $\Xi$  with the fixed transition function  $P(\cdot, \cdot | Q)$ . It is the noise process starting from  $n$   
 881 if  $\hat{Q}$  stayed constant forever:  $\{\xi_{n+j,c}(\hat{Q}), j \geq 0, \xi_{n,c}(\hat{Q}) = \xi_{n,c}\}$ . To verify A 5.4, we need to  
 882 prove for any compact set  $A \in \Xi$ ,

$$883 \quad (2.6) \quad \lim_{n,m} \frac{1}{m} \sum_{l=n}^{n+m-1} \mathbb{E}_n^Y [\xi_{l,c}(\hat{Q}) \mathbb{I}_{\{\xi_{l,c} \in A\}}] = 0$$

884 We define the corresponding fixed- $Q$  chain  $\tilde{d}_{n+j}(s^{-i}, Q)$ , for all  $j \geq 0$ , where we write  
 885  $P_{j=1}^M := P^Q$ :  

$$j \neq i$$

$$886 \quad (2.7) \quad \tilde{d}_n = d_n, \quad \tilde{d}_{k+1} = \tilde{d}_k \cdot P \prod_{j=1}^M \phi(Q^j) = \tilde{d}_k \cdot P^Q$$

$$j \neq i$$

Switching to vector notation, we write  $R_\pi$  the vector of size  $|S|$  of reward expectations  
 under the global policy  $\pi$  for the global state  $s$ . So for all  $s \in S$ ,

$$R_{\pi_k}(s) = [\phi(Q_k^{-i})(\cdot, s_k^{-i})]^T r((s^i, a^i)_c s_k^{-i}, \cdot)]$$

887 We also write  $\tilde{D}_l(Q)$  and  $D(Q)$  the corresponding state distribution vectors for  $\tilde{d}_l(s^{-i}, Q)$  the  
 888 fixed  $Q$ -chain starting in  $n$  as defined in (2.7) and  $d^{\phi(Q)}(s^{-i})$  the stationary distribution under  
 889 policy  $\phi(Q)$ .

Then for all  $n, m$ , and any  $\hat{Q}$  putting (2.5) into (2.6) allows us to rewrite the latter as:

$$\begin{aligned}
 \frac{1}{m} \sum_{l=n}^{n+m-1} \mathbb{E}_n^Y \left[ \xi_{l,c}(\hat{Q}) \mathbb{I}_{\{\xi_n \in A\}} \right] &= \frac{1}{m} \sum_{l=n}^{n+m-1} \left[ \left( \tilde{D}_l(\hat{Q}) - D(\hat{Q}) \right) R_{\pi_l} \right] \mathbb{I}_{\{\xi_l \in A\}} \\
 &= \frac{1}{m} \sum_{l=n}^{n+m-1} \left[ \left( (\tilde{D}_n(\hat{Q}) \Pi_{j=n}^{l-1} P^{\hat{Q}}) - D(\hat{Q}) \right) R_{\pi_l} \right] \mathbb{I}_{\{\xi_l \in A\}} \\
 &= \frac{1}{m} \sum_{l=n}^{n+m-1} \left[ \left( d_n \left( P^{\hat{Q}} \right)^{l-n} - D(\hat{Q}) \right) R_{\pi_l} \right] \mathbb{I}_{\{\xi_l \in A\}} \\
 &\leq \frac{R}{m} \sum_{l'=0}^{m-1} \left\| \left( d_n \left( P^{\hat{Q}} \right)^{l'} - D(\hat{Q}) \right) \right\|_1 \mathbb{I}_{\{\xi_{l'+n} \in A\}}
 \end{aligned}
 \tag{2.8}$$

From A 2.2, we know that the finite Markov chain representing the global state process is irreducible and aperiodic. Therefore,  $P^{\hat{Q}}$  is the transition matrix associated with an irreducible global state process over the finite state-space  $S$ , and the stationary distribution defined by  $D^{\phi(\hat{Q})}$  is its limiting state distribution. Moreover, the convergence rate is geometric [30], so that for any initial distribution  $d_n$  there exists constants  $0 < b < 1$  and  $C > 0$  such that for all  $l$ :

$$\left\| \left( d_n \left( P^{\hat{Q}} \right)^l - D(\hat{Q}) \right) \right\|_1 < C(1-b)^l$$

Therefore, together with (2.8) we have that:

$$\lim_m \lim_n \frac{1}{m} \sum_{l=n}^{n+m-1} \mathbb{E}_n^Y \left[ \xi_{l,c}(\hat{Q}) \mathbb{I}_{\{\xi_n \in A\}} \right] \leq \lim_m \frac{R}{m} \sum_{l'=0}^{m-1} C(1-b)^{l'} = \lim_m \frac{CR}{mb} = 0$$

**3. Example of A 3.1 for the Boltzmann distribution.** To give a better intuition of A 3.1, we look at a toy problem and a set of parameters under which it is satisfied. We then progressively extend the analysis to the general case.

Consider a single agent interacting in a game with 1 state and 2 actions and reward function  $r$ . We write

- $a_0, a_1$  the 2 actions available to the agent
- their associated rewards  $r_1 = r(a_1)$  and  $r_0 = r(a_0)$
- for any q-table  $Q$ , the associated policy  $\pi = \phi(Q)$  with its average award  $r_\pi = \sum_a \pi(a)r(a)$

Let the agent maintain a q-table estimate  $Q = [Q_1, Q_2] \in [-D, D]^2$ , and follow the corresponding policy  $\pi = \phi(Q)$  with average reward  $r_\pi$ . We introduce a perturbation to the q-table  $Q' = [Q'_1, Q'_2] \in [-D, D]^2$ , and call  $\pi' = \phi(Q')$  the associated new policy with average reward  $r_{\pi'}$ . Recall that under A 3.1, we look for a  $K \in (0, 1)$  such that

$$|r_\pi - r_{\pi'}| \leq K \|Q - Q'\|_\infty$$

We have:

$$\begin{aligned}
r_\pi &= \sum_a \pi(a)r(a) = \pi(a_0)r(a_0) + \pi(a_1)r(a_1) \\
&= \pi(a_0)r_0 + (1 - \pi(a_0))r_1
\end{aligned}
\tag{3.1}$$

and therefore

$$\begin{aligned}
|r_\pi - r_{\pi'}| &= |\pi(a_0)r_0 + (1 - \pi(a_0))r_1 - \pi'(a_0)r_0 - (1 - \pi'(a_0))r_1| \\
&= |(\pi(a_0) - \pi'(a_0))r_0 - (\pi(a_0) - \pi'(a_0))r_1| \\
&= |(\pi(a_0) - \pi'(a_0))(r_0 - r_1)| \\
&\leq |\pi(a_0) - \pi'(a_0)| |r_0 - r_1|
\end{aligned}
\tag{3.2}$$

Let us chose for  $\phi$  the mapping returning a Boltzmann distribution, parameterized by  $\lambda > 0$  an inverse temperature constant:

$$\pi = \phi(Q) \Rightarrow \forall i \quad \pi(a_i) = \frac{e^{\lambda Q_i}}{\sum_j e^{\lambda Q_j}} = \frac{1}{\sum_j e^{\lambda(Q_j - Q_i)}}$$

$\phi(Q)$  is the solution to the regularized argmax objective introduced by Equation (2.3) in Section 2, with  $\nu_{s^i}^i$  defined on the simplex as the negative entropy  $\sum_j Q_j \log(Q_j)$  and  $\tau = \frac{1}{\lambda}$  [15]. Moreover, note that for any  $[a, b] \subset \mathbb{R}$ :

$$\left| \frac{1}{1 + e^{\lambda b}} - \frac{1}{1 + e^{\lambda a}} \right| \leq \max_{x \in [a, b]} \frac{\lambda}{2 + e^{-\lambda x} + e^{\lambda x}} |b - a| \leq \frac{\lambda}{4} |b - a|$$

where we have used the mean value theorem for  $f : x \rightarrow \frac{1}{1 + e^{\lambda x}}$  and noted that  $|f'(x)|$  is maximized on  $\mathbb{R}$  with  $f'(0) = \frac{\lambda}{4}$ .

We can then write:

$$\begin{aligned}
|\pi(a_0) - \pi'(a_0)| &= \left| \frac{1}{1 + e^{\lambda(Q_1 - Q_0)}} - \frac{1}{1 + e^{\lambda(Q'_1 - Q'_0)}} \right| \\
&\leq \frac{\lambda}{4} |Q'_1 - Q'_0 - Q_1 + Q_0| \\
&\leq \frac{\lambda}{4} (|Q'_1 - Q_1| + |Q_0 - Q'_0|) \\
&\leq \frac{\lambda}{4} 2 \max(|Q'_1 - Q_1|, |Q_0 - Q'_0|) \\
&= \frac{\lambda}{2} \|Q' - Q\|_\infty
\end{aligned}
\tag{3.3}$$

Using (3.2) and (3.3), we have:

$$|r_\pi - r_{\pi'}| \leq \frac{\lambda}{2} \|Q' - Q\|_\infty |r_0 - r_1|$$

It follows that A 3.1 can be verified if there exists  $K \in (0, 1)$  such that

$$|r_0 - r_1| \leq \frac{2K}{\lambda}$$

Note that Theorem 3.1 requires the discount factor  $\beta$  to satisfy  $\beta \leq 1 - K$ , so a sufficient condition to apply the theorem is:

$$(3.5) \quad |r_0 - r_1| \leq \frac{2(1 - \beta)}{\lambda}$$

Intuitively, A 3.1 bounds the variation of the reward function over the action space. Since we know the reward to be bounded, (3.5) can be satisfied by scaling the reward appropriately without changing the optimal policy.

This conclusion extends to the case of an arbitrary large finite action space  $A$ . Then,

$$\begin{aligned} |r_\pi - r_{\pi'}| &= \left| \sum_a (\pi(a) - \pi'(a)) r(a) \right| \leq \sum_a |\pi(a) - \pi'(a)| r(a) \\ &\leq \max_u r(u) \|\phi(Q) - \phi(Q')\|_1 \\ &\leq \max_u r(u) \sqrt{|A|} \|\phi(Q) - \phi(Q')\|_2 \\ &\leq \lambda \sqrt{|A|} \max_u r(u) \|Q - Q'\|_2 \\ &\leq \lambda \max_u r(u) |A|^{\frac{3}{2}} \|Q - Q'\|_\infty \end{aligned}$$

where in the second to last line we have used the fact that  $\phi$  is Lipschitz with constant  $\lambda$ , with respect to the  $L2$  norm (see [15]). Letting  $A$  be a global action space  $A = A_1 \times \dots \times A_M$ , and  $\pi$  a global policy that is the product of  $M$  local policies, we then have for any  $a = (a^1, \dots, a^M) \in A$

$$(3.6) \quad \pi(a) - \pi'(a) = \Pi_1^M \pi^i(a^i) - \Pi_1^M \pi'^i(a^i) \leq \sum_{i=1}^M (\pi^i(a^i) - \pi'^i(a^i))$$

Finally, we reintroduce partial observability over a global action space  $S$  by writing:

$$|r_\pi - r_{\pi'}| = \left| \sum_s \sum_a \left( d^\pi(s) \pi(a | s) - d^{\pi'}(s) \pi'(a | s) \right) r(s, a) \right|$$

where  $d^\pi(s) \pi(a | s)$  is the stationary distribution over the Markov chain  $\Psi(\pi)$  on state space  $S \times A$  induced by policy  $\pi$ . It has transition function  $P^{\Psi(\pi)}((s, a), (s', a')) = P(s, a, s') \pi(a' | s')$ . Let us write  $b^\pi$  this stationary distribution. Then we have

$$|r_\pi - r_{\pi'}| \leq \max_{u \in S \times A} r(u) \|b^\pi - b^{\pi'}\|_1$$

Using classical results from the perturbation theory of Markov chains to bound the term on the right [1]:

$$(3.7) \quad |r_\pi - r_{\pi'}| \leq \max_{u \in S \times A} r(u) \kappa \left\| P^{\Psi(\pi')} - P^{\Psi(\pi)} \right\|_\infty$$

where  $\kappa$  is a condition number of the Markov chain that we will define later.

Moreover, for all pairs  $(s, a), (s', a')$  in  $S \times A$  we have:

$$(3.8) \quad P^{\Psi(\pi')}((s, a), (s', a')) - P^{\Psi(\pi)}((s, a), (s', a')) = P(s, a, s') [\pi'(a' | s') - \pi(a' | s')]$$

We write  $A_{\#} = |A_1| = \dots = |A_M|$  assuming all local action spaces are homogeneous.

Plugging (3.8) in (3.7) and then using (3.6), we have:

$$\begin{aligned} |r_{\pi} - r_{\pi'}| &\leq \kappa \max_{u \in S \times A} r(u) \max_{s, a \in S \times A} \sum_{s' \in S} \sum_{a' \in A} P(s, a, s') |\pi'(a' | s') - \pi(a' | s')| \\ &\leq \kappa \max_{u \in S \times A} r(u) \max_{s, a \in S \times A} \sum_{s' \in S} P(s, a, s') \sum_{a' \in A} \left| \sum_{i=1}^M (\pi^i(a' | s') - \pi'^i(a' | s')) \right| \\ &= \kappa \max_{u \in S \times A} r(u) \max_{s, a \in S \times A} \sum_{s' \in S} P(s, a, s') \sum_{i=1}^M \sum_{a' \in A_i} |A_{\#}|^{M-1} |\pi^i(a' | s') - \pi'^i(a' | s')| \\ &\leq \kappa \max_{u \in S \times A} r(u) |A_{\#}|^{M-1} \max_{s, a \in S \times A} \sum_{s' \in S} P(s, a, s') \sum_{i=1}^M \lambda |A_{\#}|^{\frac{3}{2}} \|Q^i(s, \cdot) - Q'^i(s, \cdot)\|_{\infty} \\ &\leq \kappa \lambda \max_{u \in S \times A} r(u) |A_{\#}|^{M+\frac{1}{2}} \sum_{i=1}^M \|Q^i - Q'^i\|_{\infty} \max_{s, a \in S \times A} \sum_{s' \in S} P(s, a, s') \\ &= \kappa \lambda \max_{u \in S \times A} r(u) |A_{\#}|^{M+\frac{1}{2}} \sum_{i=1}^M \|Q^i - Q'^i\|_{\infty} \end{aligned}$$

Several condition numbers exist for finite, irreducible Markov chains [12].  $\kappa$  can be defined as a function of the ergodic coefficient  $\tau_1$  [18]:

$$\kappa = \frac{1}{1 - \tau_1(P^{\Psi(\pi)})} \quad \tau(P) := \sup_{\substack{\|v\|_1=1 \\ v^T e=0}} \|v^T P\|_1 = \frac{1}{2} \max_{i,j} \sum_k |P_{ik} - P_{jk}|$$

Note that  $0 \leq \tau_1(P^{\Psi(\pi)}) \leq 1$  and therefore  $\kappa \geq 1$ . We have:

$$\begin{aligned} \tau_1(P^{\Psi(\pi)}) &= \frac{1}{2} \max_{(s,a)_i, (s,a)_j} \sum_{s', a'} |P((s, a)_i, s') \pi(a' | s') - P((s, a)_j, s') \pi(a' | s')| \\ &= \frac{1}{2} \max_{(s,a)_i, (s,a)_j} \sum_{s', a'} \pi(a' | s') |P((s, a)_i, s') - P((s, a)_j, s')| \\ &= \frac{1}{2} \max_{(s,a)_i, (s,a)_j} \sum_{s'} |P((s, a)_i, s') - P((s, a)_j, s')| \sum_{a'} \pi(a' | s') \\ &= \frac{1}{2} \max_{(s,a)_i, (s,a)_j} \sum_{s'} |P((s, a)_i, s') - P((s, a)_j, s')| \end{aligned}$$

We therefore define  $\kappa$  as:

$$(3.9) \quad \kappa := \frac{1}{1 - \frac{1}{2} \max_{(s,a)_i, (s,a)_j} \sum_{s'} |P((s, a)_i, s') - P((s, a)_j, s')|}$$

To conclude, when  $\phi$  is the Boltzmann distribution, Assumption A 3.1 can be verified if for any agent  $i$  and  $Q'^{\leq i}$  a perturbation of  $Q^{\leq i}$ :

$$\forall j < i, \quad \|Q^j - Q'^j\|_\infty = 0 \quad \Rightarrow \quad \forall j > i, \quad \|Q^j - Z^{\geq i+1}(Q^{< i}, Q'^i)\|_\infty \leq \|Q^i - Q'^i\|_\infty$$

Then, the necessary bound on the problem's reward becomes

$$\begin{aligned} & \kappa \lambda \max_{u \in S \times A} r(u) |A_\#|^{M+2} M \|Q^i - Q'^i\|_\infty \leq 1 - \beta \\ \Rightarrow & \max_{u \in S \times A} r(u) \leq \frac{1 - \beta}{\kappa \lambda |A_\#|^{M+\frac{1}{2}} M} \end{aligned}$$

Again, this can always be satisfied by rescaling  $r$ , although of course as the right-hand term decreases with the number of agents and available actions, rescaling will pose numerical stability issue. In practice however, as we show in our experiments in Section 4, good empirical results can be obtained without rescaling.

#### 4. Proofs of Networked multi-scale Q-learning iterates.

**4.1. Convergence of NetworkMQL.** We detail the proof of Theorem 3.2. We recall here the two conditions on the ranking of agents mentioned in Subsection 3.2:

- (A) For any node  $i$ , nodes in  $\mathcal{NA}(i)$  have a strictly inferior rank, and nodes in  $\mathcal{ND}(i)$  have a strictly superior rank.
- (B) For any node  $i$ , there exists no two different nodes of the same rank in  $\mathcal{NA}(i)$ .

We first show that any set of learning rate sequences satisfying (A) and (B) preserves the convergence of Theorem 5.2. We then discuss how learning rates belonging to that set can be derived.

*Proof.* We note that for a node  $i$ , a best response  $\pi^{*i}$  to other policies  $\pi^{-i}$  only depends on policies in the neighborhood. Any change at a local policy outside of the neighborhood does not lead to a change in best-response to  $\pi^{-i}$  if local policies in the neighborhood remain the same.

That is, let  $i$  be a node in the direct acyclic graph  $\mathcal{G}$ , and  $\pi^i$  the local policy followed by agent  $i$ . We remind that  $\mathcal{N}^i$  is the set of the neighbors of  $i$  and write the corresponding set of local policies  $\pi^{\mathcal{N}^i} = \{\pi^j, j \in \mathcal{N}^i\}$ . Similarly, we write the set of the remaining nodes excluding  $i$ ,  $\bar{\mathcal{N}}^i = \mathcal{V} \setminus \mathcal{N}^i$  and their local policies  $\pi^{\bar{\mathcal{N}}^i}$ . Let local policy  $\pi^{*i}$  be a best response to  $\pi^{-i} = (\pi^{\mathcal{N}^i}, \pi^{\bar{\mathcal{N}}^i})$ . Then, for any other set of local policies  $\pi'^{\mathcal{N}^i}$ ,  $\pi^{*i}$  is also a best response to  $\pi'^{-i} = (\pi'^{\mathcal{N}^i}, \pi^{\bar{\mathcal{N}}^i})$ . This is immediate if we recall that for  $\pi^{*i}$  is a best response to  $\pi$  if for any local policy  $\pi^i$  we have for all  $s^i \in S_i$  and  $a^i \in A_i$ ,  $Q_{\pi^{*i}}^{\pi^{-i}}(s^i, a^i) \geq Q_{\pi^i}^{\pi^{-i}}(s^i, a^i)$ , and that due to the structure of the reward as a sum of local components:

$$Q_{\pi^i}^{\pi^{-i}}(s^i, a^i) = \sum_{l=1}^M \mathbb{E}_{\pi^l, \pi^{U^l}} G(l, U^l)$$

with

$$\mathbb{E}_{(\pi^l, \pi^{U^l})} G(l, U^l) = \mathbb{E}_{(s_0^l, s_0^{U^l}, a_k^l, a_k^{U^l}, s_k) \sim (d^{\pi^l}, d^{\pi^{U^l}}, \pi^l, \pi^{U^l}, P)} \left[ \sum_{k=0}^{\infty} \beta^k r^l(s_k^l, a_k^l, s_k^{U^l}, a_k^{U^l}) \mid s_0^l = s^i, a_0^l = a^i \right]$$

and

$$Q_{\pi^i}^{\pi^{-i}}(s^i, a^i) = \sum_{l \in (i, \mathcal{N}^i)} \mathbb{E}_{(\pi^l, \pi^{U^l})} G(l, U^l) + \sum_{l \in \mathcal{N}^i} \mathbb{E}_{(\pi^l, \pi^{U^l})} G(l, U^l)$$

Now let local policy  $\pi^{*i}$  be a best response to  $\pi^{\mathcal{N}^i}$  only, and take any set of local policies excluding  $i$   $\pi^{-i} = (\pi^{\mathcal{N}^i}, \pi^{\mathcal{N}^i})$ . Then  $\sum_{l \in (i, \mathcal{N}^i)} \mathbb{E}_{\pi^l, \pi^{U^l}} G(l, U^l) \geq \sum_{l \in (i, \mathcal{N}^i)} \mathbb{E}_{\pi^l, \pi^{U^l}} G(l, U^l)$  per definition of the best response. The second sum is a constant with respect to the local policy of  $i$ , and we have  $Q_{\pi^{*i}}^{\pi^{-i}}(s^i, a^i) \geq Q_{\pi^i}^{\pi^{-i}}(s^i, a^i)$ .

It is easy to see that any ranking satisfying (A) and (B) also has the following property: there exists no two paths from two different agents of the same level towards the same node  $i$ . We show that from this property the convergence result will follow.

At each  $i$ , iterates for agents  $i$  will follow an ODE whose mean field will be a continuous function of both the parameters of agents in  $\mathcal{N}_{in}^i$ , which by construction learn on a slower scale, and the parameters of agents in  $\mathcal{N}_{out}^i$ , which by construction learn on a faster scale. At the scale of  $i$ , the parameter of  $\mathcal{N}_{in}^i$  can be considered fixed. The parameters of  $\mathcal{N}_{out}^i$  will have converged towards the parents of  $\mathcal{N}_{out}^i$ . If the  $\mathcal{G}$  is a tree, then the set of parents of  $\mathcal{N}_{out}^i$  is reduced to  $\mathcal{N}_{in}^i$ , and convergence follows. If  $\mathcal{G}$  is any standard DAG, then the mean field of any agent  $j \in \mathcal{N}_{out}^i$  is  $\propto \bar{F}^j(q^{U^j})$ . Any node  $l \in U^j$  is either slower than  $i$ , in which case it can be considered constant, or faster than  $i$ , in which case it has converged towards a continuous function of parameters  $q^{U^l}$ . Again, any node  $k$  in  $U^l$  is either slower than  $i$ , or has converged towards a continuous function of parameters  $q^{U^k}$ . By repeating this procedure, we find that  $\bar{F}^j(q^{U^j})$  depends on all the ancestors of  $j$ . Recall that the graph is acyclic, and we have just assumed that there never exists two paths from two agents at the same level towards any node. Therefore we can rewrite  $\bar{F}^j(q^{U^j})$  as a function of a finite number of agents, in which any node that has the same level with  $i$  must be  $i$  itself. Therefore, at the timescale  $i$ ,  $j$  will have converged towards a continuous function of  $q^i$  and a set of parameters slower than  $i$ . Convergence of the iterates of  $i$  towards the best response follows. This concludes the proof. ■

**Learning rates attribution procedures.** We now propose a simple algorithm that creates learning rate sequences satisfying conditions (A) and (B). Let us call **TopSort** any topological sorting algorithm. We call **RBFS** the procedure that does reverse breadth first search on the graph, starting from a single leaf up to all discovered roots, and returns the set of all visited nodes.

The detail of this algorithm is quite straightforward. First, Line 3 a total order on all nodes is extracted through a topological sort. As noted earlier this is already a valid ranking for any DAG structure that will ensure convergence, and the rest of the algorithm is dedicated to reduce the number of ranks if possible.

Lines 4 to 9 simply extract constraint sets from the DAG: two nodes belonging to the same set admit at least a path towards a same third node, and therefore cannot receive the same rank. This is done by returning, for every leaf, the set of all its ancestors through the reverse breadth first search.

Then, lines 11 to 21 simply adjust the ranking by extracting node by topological order, and assigning them the slowest rank that does not conflict with other nodes in his constraint sets. The final ranking, by construction, ensures that all nodes have higher ranks than their parents (through the topological order) and different ranks from other nodes in their constraint sets.

**Algorithm 4.1** Procedure: Attribute learning rates for any DAG structure**Require:** Network DAG  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  with  $M$  nodes

```

1: Init  $\Sigma \leftarrow \{\}, d \leftarrow [\infty, \dots, \infty] \in (\mathbb{N} \cup \infty)^M$ , Heap  $Q$ 
2:  $\mathcal{V}_R \leftarrow \text{Leaves}(\mathcal{G})$ 
3:  $Q \leftarrow \text{TopSort}(\mathcal{G})$ 
4: ## Retrieve constraint sets  $\Sigma$ 
5: for leaf in  $\mathcal{V}_R$  do
6:    $A \leftarrow \text{RDFS}(\text{leaf})$ 
7:    $\sigma \leftarrow A \cup \{\text{leaf}\}$ 
8:    $\Sigma.\text{append}(\sigma)$ 
9: end for
10: ## Assign levels based on  $\Sigma$ 
11: while  $Q$  not empty do
12:    $v \leftarrow Q.\text{pop}()$ 
13:   if  $\text{Parents}(v)$  empty then
14:      $d^v \leftarrow 0$ 
15:   else
16:      $d^v \leftarrow \max(\{d^p : p \in \text{Parents}(v)\}) + 1$ 
17:   end if
18:    $\bar{C} \leftarrow \{d^n : \exists \sigma \in \Sigma, \sigma \supset \{n, p\}\}$ 
19:   while  $d^v \in \bar{C}$  do
20:      $d^v \leftarrow d^v + 1$ 
21:   end while
22: end while return  $d$ 

```

**4.2. Extension of NetworkMQL to a ranking only satisfying hypothesis (A).** In the following, we give a sketch of the proof of Theorem 3.2, showing that convergence to the best-response equilibrium can be preserved for rankings that only satisfy hypothesis (A). We recall here this condition on the ranking of agents mentioned in Subsection 3.2:

**(A)** For any node  $i$ , nodes in  $\mathcal{NA}(i)$  have a strictly inferior rank, and nodes in  $\mathcal{ND}(i)$  have a strictly superior rank.

Any set of learning rate sequences satisfying (A) can be shown to preserve the convergence of Theorem 3.1 under new contraction assumption A 3.3. Let  $\tilde{rk} : i \rightarrow \tilde{rk}(i) \in \{1, \dots, \tilde{M}\}$  be the ranking satisfying (A) by attributing  $\tilde{M}$  different levels to the  $M$  agents. We write  $|\tilde{rk}(i)|$  the number of agents at level  $\tilde{rk}(i)$ .

The proof can be done in 4 steps:

1. First, note that the  $M$  Network MQL iterates (3.3) under ranking  $\tilde{rk}$  are equivalent to  $\tilde{M}$  multi-scale iterates of Theorem 5.1, where every iterate is updating all the agents at a same rank. Here, instead of selecting one component of the updated parameters, each iterate  $i \in \{1, \dots, \tilde{M}\}$  updates  $|\tilde{rk}(i)|$  local state-action pairs at each real time-scales.
2. Then, note that under our ranking, no two agents on the same level can be neighbors

in the interaction graph. The proof of this statement is exactly identical to the one done in the proof of Theorem 3.2 in Subsection 4.1, as it is a direct consequence of hypothesis (A) only. Just like in Subsection 4.1, this means that the associated ODE approximations for agents updated by the same iterate can be written exclusively as functions of agents learning at different levels.

3. Finally, the associated contraction property for the ODE approximations must stand when the q-tables of all agents learning at the same level are perturbed together. This is the claim made by A 3.3.

4. We conclude that by applying Theorem 5.1 under A 3.2, the convergence towards the best-response equilibrium is preserved.

## 5. Conditions for the weak convergence of multi-scale iterates with Markovian noise.

**5.1. Conditions for the weak convergence of synchronous multi-scale iterates with Markovian noise testb.** Weak convergence of stochastic approximation for two time-scales systems were proven in [21]. We formally extend these results to the multi-scale case. We consider the constrained case: at each iteration, the iterates are projected on a defined admissible space  $H$ . We assume that  $H$  is a hyperrectangle  $H = [h_1, b_1] \times [h_2, b_2] \times \cdots \times [h_d, b_d]$  with  $(h_i, b_i) \in \mathbb{R}^2$  for  $i \in \{1, \dots, d\}$  and  $d > 0$  the dimension of the iterates. The operator  $\Pi_H$  is used to denote this projection on  $H$ .

Consider  $M$  interdependent stochastic approximation processes  $\theta_k^1, \dots, \theta_k^M$  updated according to iterates:

$$(5.1) \quad \theta_{k+1}^i = \Pi_H [\theta_k^i + \alpha_k^i Y_k^i] = \theta_k^i + \alpha_k^i (F^i(\theta_k, \xi_k^i) + \delta U_k^i) + B_k^i$$

where  $\theta_k = (\theta_k^1, \dots, \theta_k^M)$ ,  $\{\xi_k^i\}$  are noise sequences,  $F^i(\cdot, \cdot)$  are functions of  $\theta$  and  $\xi^i$ ,  $\delta U_{k+1}^i = Y_k^i - F^i(\theta_k, \xi_k^i)$  are martingale noise differences,  $\alpha_k^i := \alpha^i(k) > 0$  are learning rates for timescale  $i$  at iterate  $k$ , and  $B_k^i$  is a correction term to project the iterate on  $H$ , henceforth referred as reflection terms.

Let  $\{\mathcal{F}_k\}$  be a sequence of non-decreasing  $\sigma$ -algebra generated by  $\{\theta_j^i, Y_{j-1}^i, \xi_j^i, j \leq k, i \leq M\}$ , and  $\mathbb{E}_k$  refers to the associated conditional expectation  $\mathbb{E}[\cdot | \mathcal{F}_k]$ , and we have  $\mathbb{E}_k Y_k^i = F^i(\theta_k, \xi_k^i)$ . To be concise, we will use the notations

$$\theta^{<i} := (\theta^1, \dots, \theta^{i-1}), \quad \theta^{\geq i} := (\theta^i, \dots, \theta^M).$$

We now lay down the assumptions needed to ensure convergence. Let  $\Xi$  be a complete and separable metric space, and  $A$  be an arbitrary compact set in  $\Xi$ . We start by standard assumptions for stochastic approximation algorithms: the sequences of observations  $Y_k^i$  are uniformly integrable, and at each timestep their expectations are given by a continuous function of the iterate  $\theta_k^i$ . The main idea is that an error term  $\delta U_k^i$  of null expectation will be averaged out through the iterations, so that as  $k$  goes to infinity, the behavior of the iterates can be described without the error terms. We make the following assumption for  $i = 1, \dots, M$ .

We first start with basic assumptions from stochastic approximation theory:

**A 5.1.** *The  $\{Y_k^i\}$  are uniformly integrable, and can be written  $Y_k^i = F^i(\theta_k, \xi_k^i) + \delta U_k^i$  with  $\{\delta U_k^i\}_k$  martingale noise differences  $\mathbb{E}_k \delta U_k^i = 0$  and  $F^i(\cdot, \xi^i)$  functions continuous in  $\theta$ , and continuous in  $\xi^i \in A$ .*

Here,  $F^i$  is still dependent on the error sequences  $\xi_k^i$  whose expectations are not null. Yet, the Markovian property of these sequences, combined with a constraint on the rate of evolution of the learning rates (see A 5.5), can be exploited to construct an approximation of  $F^i(\cdot, \xi^i)$  that does not depend on  $\xi^i$ . We detail some assumptions on the Markovian noise processes that can be considered.

A 5.2. *The noise processes  $\{\xi_k^i\}$  are bounded with values in  $\Xi$ , and Markovian: they admit a transition function  $P^i(\cdot, \cdot | \theta)$  such that  $P^i(\cdot, A | \theta)$  is measurable for each Borel set  $A \subset \Xi$ , and  $P^i(\xi_{k+1}^i \in \cdot | \mathcal{F}_k) = P^i(\xi_k^i, \cdot | \theta_k)$ . This transition function is continuous and does not depend on  $k$ . For any compact  $A \in \Xi$  and  $\mu \in (0, 1)$  such that, there exists a compact  $A'$  such that  $P(\xi_{k+1}^i \in A' | \xi_k^i) \geq 1 - \mu$  for all  $\xi_k^i \in A$ .*

We now define the fixed  $\theta$ -chain  $\{\xi_k(\theta)\}$ , the Markov chain on state space  $\Xi$  with the fixed transition function  $P(\xi, \cdot, | \theta)$ . It is the noise process starting from  $k$  if  $\theta$  stayed constant, i.e.,  $\{\xi_{k+j}(\theta), j \geq 0, \xi_k(\theta) = \xi_k\}$ . The continuous function of the actual noise process can be approximated by the continuous function of the fixed-chain process  $F^i(\cdot, \xi_k^i)$  (see the proof of Theorem 8.4.3 [21], p.271-275) if the rate of change of the learning rates is slow enough. If we can construct a function  $\hat{F}^i(\cdot)$  of  $\theta$  that does not depend on the process  $\xi^i$ , such that  $\hat{F}^i(\theta)$  is a local average of the  $F^i(\cdot, \xi_k^i)$ , then  $\hat{F}^i(\theta_k)$  is also an approximation of  $F^i(\theta_k, \xi^i)$  as  $k \rightarrow \infty$ . We detail these assumptions here:

A 5.3. *The set  $\{F^i(\theta_k, \xi_k^i)\}_k$  is uniformly integrable. For any  $j \geq 0$  with  $\xi_j^i \in A$  the set  $\{F^i(\theta, \xi_{j+k}^i(\theta))\}_{k \geq 0}$  is uniformly integrable, where  $\xi_{j+k}^i(\theta)$  is the fixed- $\theta$  chain with initial conditions  $\xi_j^i$  and transition function  $P^i(\xi, \cdot | \theta_j)$ .*

A 5.4 (Averaging condition). *There exists a continuous function  $\bar{F}^i(\cdot)$  such that for each  $\theta$  and on any compact set  $A \in \Xi$*

$$\lim_{(k,m) \rightarrow \infty} \frac{1}{m} \sum_{j=k}^{k+m-1} \mathbb{E}_k [F^i(\theta, \xi_j^i(\theta)) - \bar{F}^i(\theta)] I_{\xi_k^i \in A} = 0$$

We establish a different timescale to correspond to each process. For  $i, j \in \{1, \dots, M\}$ , let  $t_k^j = \sum_{l=0}^{k-1} \alpha_l^j$ , and  $\theta_{\alpha^j}^{i,0}(t)$  be the piecewise interpolation of the process  $\theta_k^i$  on the  $j$ -th timescale defined as

$$\theta_{\alpha^j}^{i,0}(t) = \theta_0^i, \quad t \leq 0, \quad \theta_{\alpha^j}^{i,0}(t) = \theta_k^i, \quad t \in [t_k^j, t_{k+1}^j]$$

Then, the shifted continuous time interpolation  $\theta_{\alpha^j}^{i,k}(\cdot)$  is simply the interpolation "started" from a specific time-step  $k$ :

$$(5.2) \quad \theta_{\alpha^j}^{i,k}(t) = \theta_{\alpha^j}^{i,0}(t_k^j + t)$$

and we let  $m^{(j)}(t) = \{\kappa : t_\kappa^j \leq t \leq t_{\kappa+1}^j\}$ . Similarly, we define  $B_{\alpha^j}^{i,k}$  the shifted continuous time interpolation at the  $j$ -th timescale of the sequence of reflection terms  $B_k^i$ . We are interested in the behavior of  $\theta_{\alpha^j}^{i,k}(\cdot)$  and  $B_{\alpha^j}^{i,k}(\cdot)$  as  $t_k^j \rightarrow \infty$  while  $\alpha_k^j \rightarrow 0$ . We now lay out further constraints on the learning rate sequences. The first two are standard for the stochastic approximating

literature: intuitively they require the learning rates to go towards zero, but not too quickly. The third assumption is what makes the iterates multi-scale: it imposes a hierarchy between the  $M$  sequences that ensures every iterate is learning at a different timescale.

A 5.5 (Assumption on learning rates). For each  $i \in \{1, \dots, M\}$ ,

- (Classical rates)  $\lim_k \alpha_k^i = 0$  and  $\sum_{k=0}^{\infty} \alpha_k^i = \infty$
- (Slow changes) there is a sequence of integers  $a_n^i \rightarrow \infty$  such that

$$\lim_n \sup_{0 \leq j \leq a_n^i} \left| \frac{\alpha_{n+j}^i}{\alpha_n^i} - 1 \right| = 0$$

- (Multi-scale)  $\frac{\alpha_k^i}{\alpha_k^j} \rightarrow 0$ , as  $k \rightarrow \infty$ , whenever each  $i < j$ .

With the expectations  $\mathbb{E}_k Y_k^i$  being approximated by  $\bar{F}^i(\theta_k)$  as  $k$  goes to  $\infty$ , the interpolations of the iterates  $\theta_{\alpha_j^i}^{i,k}$  will be shown to admit limit processes following mean ODEs defined by the  $\bar{F}^i$ . The solution of the ODE can then be used to characterize the asymptotic properties of the  $\theta_k^i$  for  $i = 1, \dots, M$ . Thanks to the multi-scale assumption, at any timescale  $j$  the interpolation for all iterates learning at a slower timescale  $i < j$  will follow the null ODE. Intuitively, they evolve so slowly that they can be considered constant at the  $j$ -th timescale. Similarly, the interpolations for all iterates learning at a faster timescale can be considered to have reached the limit of their respective mean ODE, if it exists. We consider the case where the ODE for every limit process for any timescale has a unique asymptotically stable point.

A 5.6. There exists a continuous function  $\zeta^i(\theta^{<i})$  such that, for any set of initial conditions  $\theta$ , the solution to the following ODE has a unique asymptotically stable point  $(\theta^{<i}, \zeta^i(\theta^{<i}))$  for  $i \geq 2$ :

$$\begin{aligned} \dot{X}^j &= 0 \quad \text{for } j < i \\ \dot{X}^i &= \bar{F}^i(X^{<i+1}, Z^{\geq i+1}(X^{<i+1})) + b^i. \end{aligned}$$

where  $b^i$  is the reflection on  $H$ , and

$$(5.3) \quad Z^{\geq i}(\theta_k^{<i}) = (\zeta^i(\theta_k^{<i}), Z^{\geq i+1}(\theta_k^{<i}, \zeta^i(\theta_k^{<i}))), \quad i = 2, \dots, M-2$$

with  $Z^{\geq M-1}(\theta_k^{<M-1}) = (\zeta^{M-1}(\theta_k^{<M-1}), \zeta^M(\theta_k^{<M-1}, \zeta^{M-1}(\theta_k^{<M-1})))$ .

Note that the reflection terms  $b^i$  of the projected ODE must live within a convex space  $\Upsilon(X^i)$ , defined the following way: on the interior of  $H$ ,  $\Upsilon(X^i) = \{0\}$ , the set only containing the null vector, and on the boundary of  $H$ ,  $\Upsilon(X^i)$  is the infinite convex cone generated by the outer normals at  $X^i$  of the faces on  $H$  on which  $X^i$  lies.

Now we state the weak convergence of the iterates (5.1) in the following theorem.

Theorem 5.1 (Weak convergence of multi-scale iterates with Markovian noise). Consider iterates (5.1). Let  $\{\theta_{\alpha_j^i}^{i,k}(\cdot)\}$  be the interpolation of the process  $\theta_k^i$  on the  $j$ -th timescale, defined by (5.2). If ?? 5.1–5.5 hold, then  $\{\theta_{\alpha_1^1}^{1,k}(\cdot)\}$  admits a subsequence which converges towards a process  $\theta^1(\cdot)$  such that:

$$(5.4) \quad \dot{\theta}^1 = \bar{F}^1(\theta^1, Z^{\geq 2}(\theta^1)) + b^1, \quad b^1(t) \in -\Upsilon(\theta^1(t))$$

1133 where  $b^1$  is the reflection, that is the minimum force needed to keep  $\theta^1$  in  $H$ . Moreover, for  
 1134 any  $\delta > 0$ , the fraction of time spent by  $\theta^1(\cdot)$  in any  $\delta$ -neighborhood around the set of limit  
 1135 points of (5.4) on the interval  $[0, T]$  goes to one in probability as  $T \rightarrow \infty$ .

1136 Theorem 5.1 is a straightforward extension of the weak convergence result established  
 1137 for two timescale iterates by [Theorem 8.6.1, [21], p.286] to the case of  $M$  time scales. An  
 1138 example of the extension procedure from two-scale to multi-scale can be found in [22]. The  
 1139 idea behind Kushner's original proof in [21] for the two-timescale case is that the noise induced  
 1140 by the Markovian sequences  $\{\xi_k^i\}$  can be seen as perturbations to local averages defined by  
 1141 the functions  $\bar{F}^i$ . This allows to approximate the iterates in continuous time by a projected  
 1142 ODE.

1143 **5.2. Extension to asynchronous iterates.** We will now consider the case where the it-  
 1144 erates are updated asynchronously: that is, not all elements of the  $\theta^i$  are updated at every  
 1145 iteration.

1146 We index all elements in every  $\theta^i$  by  $c \in \{1 \dots C\}$ , and the  $C$  elements are updated in an  
 1147 asynchronous manner. Let  $\alpha_{k,c}^i$  be the learning rate for element  $c$  of iterate  $i$  at timestep  $k$ :  
 1148 all elements within a single iterate are given the same sequences of learning rates, so that we  
 1149 use the notation  $\alpha_k^i = \alpha_{k,1}^i = \alpha_{k,2}^i = \dots = \alpha_{k,C}^i$ . The  $M$  iterates in (5.1) can therefore be seen  
 1150 as  $M \times C$  iterates, with the updates to each component following:

$$1151 \quad (5.5) \quad \theta_{k+1,c}^i = \Pi_H [\theta_{k,c}^i + \alpha_{k,c}^i Y_{k,c}^i]$$

1152 The time between the  $k$ th and  $(k+1)$ th updates of the element indexed by  $c$  in  $\{\theta_k^i\}_k$  is  
 1153 given by the random variable  $\tau_{k,c}^i$ . Because the  $k$ th update can happen at a different time for  
 1154 two components, we need another timeline to analyze the behavior of the iterates. We will  
 1155 look at their behaviors in the "real time", so that the  $k$ th update at element  $c$  in the  $\{\theta_k^i\}_k$  is  
 1156 done at the real time  $T_{k,c}^i = \sum_{n=0}^{k-1} \tau_{n,c}^i$ . We note  $\Gamma_{k,c}^i = \sum_{n=0}^{k-1} \alpha_n^i \tau_{n,c}^i$  the corresponding scaled  
 1157 real time, and introduce the real-time interpolation  $\hat{\theta}_c^i: \hat{\theta}_c^i(t) = \theta_{k,c}^i$  on  $[T_{k,c}^i, T_{k+1,c}^i)$ .

Like in (5.2), we look at the shifted piecewise constant interpolations  $\theta_{c,\alpha^j}^i$  of the sequences  
 $\{\theta_{k,c}^i\}_k$  at every timestep  $j = \{1, \dots, C\}$  in the iterate time, that is the continuous interpo-  
 lations whose origins are at any arbitrary timestep  $k$ . Here again, since all components do  
 not reach a given timestep at the same time, we define the shifted interpolates as starting at  
 arbitrary real times  $v$ . For this purpose, we introduce functions  $p_c^i(v)$ , that return the index  
 of the first update at an element  $c$  of iterate  $i$  after a given real time  $v$ :

$$p_c^i(v) = \min \left\{ k : \sum_{n=0}^{k-1} \tau_{n,c}^i \geq v \right\}, \quad \forall v > 0,$$

1158 The shifted interpolates are then

$$1159 \quad (5.6) \quad \theta_{c,\alpha^j}^{i,v}(t) = \theta_{k+p_c^i(v),c}^i, \quad t \in [t_{k,c}^{ij,v}, t_{k+1,c}^{ij,v}), \quad t_{k,c}^{ij,v} = \sum_{n=p_c^i(v)}^{k-1} \alpha_n^j$$

1160 and the shifted real-time interpolations  $\theta_{c,\alpha^j}^{i,v}(\cdot)$  are defined similarly:

$$1161 \quad (5.7) \quad \hat{\theta}_{c,\alpha^j}^{i,v}(t) = \theta_{k,c}^i, \quad t \in [\Gamma_{k+p_c^i,c}^{ij,v}, \Gamma_{k+1,c}^{ij,v}) \quad \Gamma_{k,c}^{ij,v} = \sum_{n=p_c^i(v)}^{k-1} \alpha_n^j \tau_{n,c}^i$$

1162 We now extend the definitions of the  $\sigma$ -algebra used in Subsection 5.1. Two sets of random  
 1163 variables need to be considered at every iteration: the  $Y_{k,c}^i$  and the  $\tau_{k+1,c}^i$ . The corresponding  
 1164  $\sigma$ -algebras should measure all variables observed in the "past" up to the relevant moment  
 1165 during update  $k+1$ . Again reasoning in real time, note that update  $k+1$  is made after  
 1166 having observed  $Y_k^i$ , but before entering the next waiting time  $\tau_{k+1,c}^i$ . This corresponds to two  
 1167 slightly different sequences of  $\sigma$ -algebras:

$$1168 \quad \begin{aligned} \mathcal{F}_{k,c}^{i,\tau} &= \{\theta_{0,c}^i, Y_{j-1,h}^i, \xi_{j-1,h}^i, \tau_{j-1,h}^i \mid T_{j,h}^i \leq T_{k+1,c}^i\} \\ \mathcal{F}_{k,c}^{i,Y} &= \{\theta_{0,c}^i, Y_{j-1,h}^i, \xi_{j-1,h}^i \mid T_{j,h}^i < T_{k+1,c}^i\} \cup \{\tau_{j-1,h}^i \mid T_{j,h}^i \leq T_{k+1,c}^i\} \end{aligned}$$

1169 We write the associated conditional expectations  $\mathbb{E}_{k,c}^{i,\tau}$  and  $\mathbb{E}_{k,c}^{i,Y}$ .

1170 Let us denote the component-wise error sequences  $\xi_{k,c}^i$ ,  $\delta U_{k,c}^i$ ,  $\xi_k^i = (\xi_{k,1}^i, \dots, \xi_{k,C}^i)$ , and  
 1171  $\delta U_k^i = (\delta U_{k,1}^i, \dots, \delta U_{k,C}^i)$ . We assume ?? 5.1–5.5 hold, with any statement on a sequence  
 1172  $X_k^i$  interpreted as holding for all component-wise sequences  $X_{k,c}^i$ . We make the additional  
 1173 assumptions on the time intervals between updates:

1174 A 5.7. For all  $i$ , the sequence of intervals between updates  $\{\tau_{k,c}^i\}_k$  is uniformly integrable,  
 1175 and there exists  $\bar{u}_c^i \geq 1$  such that the  $\mathbb{E}_{k+1,c}^{i,\tau} [\tau_{k,c}^i]$  are in the bounded interval  $[1, \bar{u}_c^i]$  uniformly  
 1176 in  $k$ .

A 5.8. Every component's learning rate  $\alpha_{k,c}^i$  can be written as a local average of positive  
 real-valued functions  $f^i$ :

$$\alpha_{k,c}^i = \frac{1}{\tau_{k,c}} \int_{T_{k,c}^i}^{T_{k,c}^i + \tau_{k,c}^i} f^i(s) ds \quad \text{such that} \quad \int_0^\infty f^i(s) ds = \infty \quad \text{and} \quad \lim_{s \rightarrow \infty} f^i(s) = 0$$

1177 A 5.9. There exists a continuous function  $\zeta^i(\theta^{<i})$  such that, for any set of initial conditions  
 1178  $\theta$ , the solution to the following ODE has a unique asymptotically stable point  $(\theta^{<i}, \zeta^i(\theta^{<i}))$   
 1179 for  $i \geq 2$ :

$$1180 \quad \begin{aligned} \dot{X}^j &= 0 \quad \text{for } j < i \\ \dot{X}^i &= \frac{\bar{F}^i(X^{<i+1}, Z^{\geq i+1}(X^{<i+1}))}{u_c^i} + \hat{b}^i. \end{aligned}$$

1181 with  $u_c^i(t)$  with values in  $[1, \bar{u}_c^i]$ ,  $\hat{b}^i$  the term of projection on  $H$ , the  $Z^i$  have been defined in  
 1182 (5.3)

1183 Then, we can state the weak convergence result for the asynchronous multi-scale iterates.

Theorem 5.2 (Weak convergence of asynchronous multi-scale iterates with Markovian noise). Consider iterates (5.5), updated asynchronously following the time interval sequences  $\{\tau_{k,c}^i\}_k$ . If Assumptions ?? 5.1–5.5 hold, and Assumptions A 5.7 and ?? also hold, then the conclusion of Theorem 5.1 still holds with the limit process:

$$(5.8) \quad \dot{\theta}_{c,\alpha^1}^1(t) = \frac{\bar{F}^1(\theta_{c,\alpha^1}^1(t), Z^{\geq 2}(\theta_{c,\alpha^1}^1)(t))}{u_c^i(t)} + \hat{b}_{c,\alpha^1}^1(t) \quad u_c^i(t) \in [1, \bar{u}_c^i].$$

The weak convergence of asynchronous updates for the single-agent case has been established in [Theorem 12.3.5 [21]], and this is an extension to the multi-scale case. As previously for Theorem 5.1, the extension is derived by writing the ODEs for the continuous approximations at all iterate timescales. Unlike before, the ODEs are now dependent on the continuous approximation at the real timescale. A simple relation between the approximations at real and iterate timescales introduced by [21] can then be used to derive ODEs for the latter and conclude the proof.

**6. Proofs: Weak convergence of multi-scale iterates.** In the following, we formally extend the proofs of the weak convergence of synchronous two-timescale updates [Theorem 6.2, Chapter 8, [21]], and of single-timescale asynchronous updates [Theorem 5.1, Chapter 12, [21]] to prove the weak convergence of asynchronous multi-scale iterates with Markovian noise (Theorem 5.2). We first prove Theorem 5.1 for synchronous multi-scale iterates with Markovian noise, and then extend these results to the asynchronous case in Theorem 5.2.

**6.1. Proof of Theorem 5.1: Weak convergence of multi-timescale iterates.** Before proceeding to the proof of Theorem 5.1, we need the following lemmas. Lemma 6.1, shows that the continuity of the paths of any weak sense limit process will imply the existence of convergent subsequences with Lipschitz continuous paths. Then, Lemma 6.4 and Lemma 6.2 are used to show that a process is a martingale. This martingale will have Lipschitz continuous paths and hence will be proven to be a constant, leading to the mean projected ODEs. Finally, Lemma 6.3 will be used to characterize the reflection term  $b^1$  of this ODE.

The proofs of Lemma 6.1 and Lemma 6.3 can be found in [21] p.253 and p.109. The proof of Lemma 6.4 can be found in the proof of Theorem 8.4.3 [21], p.271-275.

Lemma 6.1 ([21], p.233 - Theorem 7.3.5). Let us consider any continuous process  $X(\cdot)$  in a path space  $\{X(\cdot)\}_t \in O[-\infty, \infty)$ , defined as the space of real-valued functions on the interval  $[0, \infty)$  that are right continuous and have left-sided limits. If  $X$  is of dimension  $C > 0$ , we write  $O^C(-\infty, \infty)$ . Let there be a sequence of such processes  $\{X^k(t)\}_k$ . It is said to be tight if for each  $\mu > 0$  there is a compact set  $A_\mu \in O[0, \infty)$  such that  $\inf_k P(X^k(\cdot) \in A_\mu) > 1 - \mu$  and

$$\lim_{\delta \rightarrow 0} \limsup_n \sup_{|\tau| \leq T} \sup_{s \leq \delta} \min(|X^k(\tau + s) - X^k(\tau)|, 1) = 0$$

In particular, let  $\{Y_k^n, n \geq 0, k \geq 0\}$  be a sequence of real-valued random variables that is uniformly integrable, i.e.  $\lim_{K \rightarrow \infty} \sup_n E|Y_n|I_{\{|Y_n| \geq K\}} = 0$ , and let  $\alpha_k^i$  be non-negative numbers that satisfy for all  $n$

$$\sum_{l=0}^{\infty} \alpha_l^n = \infty, \quad \limsup_n \sup_{k \geq 0} \alpha_k^n = 0.$$

Define  $t_k^n = \sum_{l=0}^{k-1} \alpha_l^n$  and the processes  $X^n(\cdot)$  on  $O^r[0, \infty)$  by

$$X^n(t) = \sum_{l=0}^{k-1} \alpha_l^n Y_l^n \quad t \in [t_k^n, t_{k+1}^n]$$

1216 Then  $\{X^n(\cdot)\}$  is tight, and all weak sense limit processes have Lipschitz continuous paths with  
1217 probability one.

Lemma 6.2 ([21], p.234 - Theorem 7.4.1). Let  $U(\cdot)$  be a random process where  $U(t)$  is measurable on the  $\sigma$ -algebra  $\mathcal{F}_t^V$  determined by  $\{V(s), s \leq t\}$  for a given process  $V(\cdot)$  and  $\mathbb{E}|U(t)| < \infty$  for each  $t$ . Suppose that for each real  $t > 0$  and  $\tau \geq 0$ , and for any integer  $p$  and each set of real numbers  $s_i \leq t, i = 1 \dots p$ , and  $h$  a bounded and continuous real-valued continuous, we have:

$$\mathbb{E} [h(V(s_i), i \leq p) [U(t + \tau) - U(t)]] = 0$$

1218 Then,  $U(\cdot)$  is a  $\mathcal{F}_t^V$ -martingale process.

1219 Lemma 6.3 ([21], p.109 - Theorem 4.3.1). Let  $H$  be an hyperrectangle constraint space.  
1220 Let there be continuous functions  $\bar{F}(\cdot)$  bounded on  $H$ ,  $x(\cdot)$  Lipschitz, and  $B(\cdot)$  such that for  
1221 any  $\tau > 0, x(t) \in H$ ,  $B(0) = 0$  and  $x(t + \tau) = x(0) + \int_t^{t+\tau} \bar{F}(x(s))ds + B(t + \tau) - B(t)$ , and  
1222 an upper semi-continuous set function  $\Upsilon$ . If the following conditions are true: ‘

- 1223 •  $B(t + \tau) - B(t) = 0$  if  $x(s) \in$  the interior of  $H$  for almost all  $s \in [t, t + \tau]$
- 1224 •  $B(t + \tau) - B(t)$  belongs to the convex hull of all  $\Upsilon(x(s))$  for  $s \in [t, t + \tau]$

1225 Then,  $B(\cdot)$  is absolutely continuous, and there is a measurable function  $b(\cdot)$  such that  $b(t) \in$   
1226  $\Upsilon(x(t))$  for almost all  $t$  and  $B(t + \tau) - B(t) = \int_t^{t+\tau} b(s)ds$

1227 Lemma 6.4. Consider iterates  $\{\theta_k^i\}$ ,  $i \in \{1, \dots\}$ , defined by (5.1). Assume that A 5.1-  
1228 A 5.4 and A 5.5 (slow changes) are true. Then for  $t > 0$  and  $\tau \geq 0$ , and for any integer  $p$  and  
1229 each set of real numbers  $s_j \leq t, j = 1 \dots p$ ,

$$1230 \quad \lim_{k \rightarrow \infty} \mathbb{E} h(\theta_{\alpha^l}^k(s_j), j \leq p) \left[ \sum_{n=m^{(l)}(t_k^l + t)}^{m^{(l)}(t_k^l + t + \tau) - 1} \alpha_n^l \left( F^l(\theta_n, \xi_n^l) - \bar{F}^l(\theta_n) \right) \right] = 0, \quad l = 2, \dots, M$$

1231 with  $h$  any bounded, continuous, and real-valued function of its arguments.

1232 For a fixed time scale  $l \in \{2, \dots, M\}$ , the stochastic approximation processes (5.1) can be  
1233 rewritten in the form, for  $2 \leq i \leq l \leq M$

$$1234 \quad \theta_{k+1}^i = \theta_k^i + \alpha_k^l \frac{\alpha_k^i}{\alpha_k^l} (F_k^i(\theta_k, \xi_k^i) + \delta U_k^i + B_k^i),$$

1235 Then, the shifted interpolations (5.2) on the  $l$ -th timescale can be rewritten in the following  
1236 way

$$1237 \quad (6.1) \quad \theta_{\alpha^l}^{i,k+1}(t) = \theta_k^i + \bar{G}_{\alpha^l}^{i,k}(t) + M_{\alpha^l}^{i,k}(t) + H_{\alpha^l}^{i,k}(t) + A_{\alpha^l}^{i,k}(t) + B_{\alpha^l}^{i,k}(t)$$

where

$$\bar{G}_{\alpha^l}^{i,k}(t) = \sum_{n=k}^{m^{(l)}(t_k^l+t)-1} \alpha_j^l \frac{\alpha_n^i}{\alpha_n^l} \bar{F}_n^i(\theta_n), \quad M_{\alpha^M}^{i,k}(t) = \sum_{n=k}^{m^{(l)}(t_k^l+t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} \delta U_n^i$$

$$B_{\alpha^l}^{i,k}(t) = \sum_{n=k}^{m^{(l)}(t_k^l+t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} B_n^i, \quad H_{\alpha^l}^{i,k}(t) = \sum_{n=k}^{m^{(l)}(t_k^l+t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} [F_n^i(\theta_n, \xi_n^i) - \bar{F}^i(\theta_n)].$$

and  $A_{\alpha^l}^{i,k}(t) = 0$  if  $i \neq l < M$  or  $l = M$ ,

$$(6.2) \quad A_{\alpha^l}^{i,k}(t) = \sum_{n=k}^{m^{(l)}(t_k^l+t)-1} \alpha_n^l \left[ \bar{F}^l(\theta_n) - \bar{F}^l(\theta_n^{<l+1}, Z^l(\theta_n^{<l})) \right], \text{ if } i = l < M$$

Note that (6.2) is the tracking error at the timescale  $l$ .

We can now define the process  $W_{\alpha^j}^{i,k}(\cdot)$  by

$$(6.3) \quad W_{\alpha^l}^{i,k}(t) = \theta_{\alpha^l}^{i,k}(t) - \theta_k^i - \bar{G}_{\alpha^l}^{i,k}(t) - B_{\alpha^l}^{i,k}(t)$$

It follows from (6.1) that  $W_{\alpha^l}^{i,k}(t) = M_{\alpha^l}^{i,k}(t) + H_{\alpha^l}^{i,k}(t) + A_{\alpha^l}^{i,k}(t)$ .

Let  $h$  be a bounded, continuous, and real-valued function of its arguments. Fix  $t > 0$  and  $\tau \geq 0$ , and for any integer  $p$  and each set of real numbers  $s_j \leq t, j = 1 \dots p$ , we have

$$(6.4) \quad 0 = \mathbb{E} \left[ h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ W_{\alpha^l}^{i,k}(t + \tau) - W_{\alpha^l}^{i,k}(t) \right] \right]$$

$$(6.5) \quad - \mathbb{E} \left[ h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ M_{\alpha^l}^{i,k}(t + \tau) - M_{\alpha^l}^{i,k}(t) \right] \right]$$

$$(6.6) \quad - \mathbb{E} \left[ h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ H_{\alpha^l}^{i,k}(t + \tau) - H_{\alpha^l}^{i,k}(t) \right] \right]$$

$$(6.7) \quad - \mathbb{E} \left[ h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ A_{\alpha^l}^{i,k}(t + \tau) - A_{\alpha^l}^{i,k}(t) \right] \right]$$

It will be shown later that terms (6.5)-(6.7) go to zero as  $k \rightarrow \infty$ . Before that, let us suppose temporarily convergences to zero of these terms, which leads to convergence to zero of  $\mathbb{E} \left[ h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^{i,k}, j \leq p) \left[ W_{\alpha^l}^{i,k}(t + \tau) - W_{\alpha^l}^{i,k}(t) \right] \right]$ .

If A 5.3 and A 5.5 are true, then according to Lemma 6.1 we have the tightness of the interpolation processes  $\{\theta_{\alpha^l}^{i,k}, B_{\alpha^l}^{i,k}\}$  and Lipschitz continuity of paths of any weak sense limit process. It follows that from any subsequence of  $\{\theta_{\alpha^l}^{i,k}, B_{\alpha^l}^{i,k}\}$  we can extract a further subsequence indexed by  $(n_k^i)$  as  $k \rightarrow \infty$  and processes  $\{\theta_{\alpha^l}^i, B_{\alpha^l}^i\}$  with Lipschitz continuous paths such that  $(\theta_{\alpha^l}^{i,n_k^i}(\cdot), B_{\alpha^l}^{i,n_k^i}(\cdot)) \rightarrow (\theta_{\alpha^l}^i(\cdot), B_{\alpha^l}^i(\cdot))$ .

Let us now characterize  $W_{\alpha^l}^i(t)$ , limits of the processes  $W_{\alpha^l}^{i,n_k^i}(t)$  as  $k \rightarrow \infty$ , with

$$(6.8) \quad W_{\alpha^l}^i(t) = \theta_{\alpha^l}^i(t) - \theta_{\alpha^l}^i(0) - \int_0^t \bar{g}_{\alpha^l}^i(\theta_{\alpha^l}^i(s)) ds - B_{\alpha^l}^i(t)$$

where

$$\bar{g}_{\alpha^l}^i(\theta_{\alpha^l}(s)) = \lim_{j \rightarrow \infty} \frac{\alpha_{n_j}^i}{\alpha_{n_j}^l} \bar{F}^i(\theta_{n_j^l}).$$

Then, the assumed convergence of the terms (6.4)-(6.7) and the weak convergence of  $(\theta_{\alpha^l}^{i,n_k}(\cdot), B_{\alpha^l}^{i,k}(\cdot))$  imply that

$$\mathbb{E}h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) [W_{\alpha^l}^i(t + \tau) - W_{\alpha^l}^i(t)] = 0$$

Since  $W_{\alpha^l}^i$  is a function of  $\{\theta_{\alpha^l}(s), s \leq t\}$  and  $h, p, t, s_i \leq t, \tau$  are arbitrarily chosen, Lemma 6.2 implies that  $W_{\alpha^l}^i$  is a continuous  $\mathcal{F}_t$ -martingale process, and

$$\mathbb{E} [W_{\alpha^l}^i(t + \tau) - W_{\alpha^l}^i(t) | \theta_{\alpha^l}(s), B_{\alpha^l}^k(s_j), s \leq t] = 0$$

1264 Since  $\theta_{\alpha^l}(\cdot)$  has Lipschitz-continuous paths with probability one,  $W_{\alpha^l}^i(\cdot)$  is a constant process.  
1265 Since  $W_{\alpha^l}^i(0) = 0, W_{\alpha^l}^i(t) = 0$  for all  $t$ . Therefore, it follows from (6.8) that

$$1266 \quad (6.9) \quad \theta_{\alpha^l}^i(t) = \theta_{\alpha^l}^i(0) + \int_0^t \bar{g}_{\alpha^l}^i(\theta_{\alpha^l}^i(s)) ds + B_{\alpha^l}^i(t)$$

It follows from A 5.4 (continuity of  $\bar{F}^i$ ) and A 5.5 (multi-scale learning rates) that

$$\bar{g}_{\alpha^l}^i(\theta_{\alpha^l}(s)) = \begin{cases} 0, & i < l, \\ \bar{F}^l(\theta_{\alpha^l}(s)) & i = l \end{cases}$$

1267 It remains to characterize  $B_{\alpha^l}^i$  with Lemma 6.3. Note that reflection terms  $B_k^i$  must live within  
1268 a convex space  $\Upsilon(\theta_k^i)$ , defined the following way: on the interior of  $H$ ,  $\Upsilon(\theta^i) = \{0\}$ , the set  
1269 only containing the null vector, and on the boundary of  $H$ ,  $\Upsilon(\theta^i)$  is the infinite convex cone  
1270 generated by the outer normals at  $\theta^i$  of the faces of  $H$  on which  $\theta^i$  lies. Therefore, on any  
1271 interval  $[t, t + \tau]$   $B_{\alpha^l}^{i,k}(t + \tau) - B_{\alpha^l}^{i,k}(t)$  belongs to the convex hull of all  $\Upsilon(s), s \in [t, t + \tau]$ . Now,  
1272 starting our iterates in the constraint space will ensure  $B_{\alpha^l}^i(0) = 0$ , and we have just shown  
1273 that  $B_{\alpha^l}^i(\cdot)$  has Lipschitz-continuous paths. We conclude that there exists a function  $b_{\alpha^l}^i(\cdot)$ ,  
1274 with  $b_{\alpha^l}^i(t) \in \Upsilon(\theta_{\alpha^l}^i)$  such that  $B_{\alpha^l}^i(t) = \int_0^t b_{\alpha^l}^i(s) ds$ . We hence have that the limit processes  
1275  $\theta_{\alpha^l}^i, i \leq l$  follows the mean ODE

$$1276 \quad (6.10) \quad \begin{aligned} \dot{\theta}_{\alpha^l}^i &= 0, & i < l \\ \dot{\theta}_{\alpha^l}^l &= \bar{F}^l(\theta_{\alpha^l}) + b_{\alpha^l}^l(s) \end{aligned}$$

1277 Now we prove the convergences to zero of terms (6.5)-(6.7). Recall that  $\mathcal{F}_{t_k^l+t}$  is the  
1278  $\sigma$ -algebra measuring all  $\{\theta_{\alpha^l}^k(s), s \leq t\}$ . Fix  $t > 0$  and  $\tau \geq 0$ , and for any integer  $p$  and each  
1279 set of real numbers  $s_j \leq t, j = 1 \dots p$ , we have

$$1280 \quad \begin{aligned} &\mathbb{E}h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) [M_{\alpha^l}^{i,k}(t + \tau) - M_{\alpha^l}^{i,k}(t)] \\ &= \mathbb{E}h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \mathbb{E}_{\mathcal{F}_{t_k^l+t}} \left[ \sum_{n=m^{(l)}(t_k^l+t)}^{m^{(l)}(t_k^l+t+\tau)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} \delta U_n^i \right] \end{aligned}$$

with  $h$  any bounded, continuous, and real-valued function of its arguments. It follows from A 5.1 (martingale difference) that the term (6.5) goes to zero as  $k \rightarrow \infty$ .

For the term (6.6), we have by definition

$$\begin{aligned} & \mathbb{E}h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ H_{\alpha^l}^{i,k}(t + \tau) - H_{\alpha^l}^{i,k}(t) \right] \\ &= \mathbb{E}h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ \alpha_n^l \sum_{n=m^{(l)}(t_k^l+t)}^{m^{(l)}(t_k^l+t+\tau)-1} (F^i(\theta_n, \xi_n^i) - \bar{F}^i(\theta_n)) \right] \end{aligned}$$

By Lemma 6.4 and A 5.1-A 5.5, convergence to zero of the term (6.6) follows.

Finally, the convergence to zero of the term (6.7) follows from

$$\begin{aligned} & \mathbb{E}h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ A^{l,k}(t + \tau) - A^{l,k}(t) \right] \\ &= \mathbb{E}h(\theta_{\alpha^l}^k(s_j), B_{\alpha^l}^k(s_j), j \leq p) \left[ \alpha_n^l \sum_{n=m^{(l)}(t_k^l+t)}^{m^{(l)}(t_k^l+t+\tau)-1} \left( \bar{F}^l(\theta_n) - \bar{F}^l(\theta_n^{\leq i+1}, Z^{\geq i+1}(\theta_n^{\leq i+1})) \right) \right] \end{aligned}$$

and A 5.4 (continuity of  $\bar{F}^i$ ).

It remains to show (5.4). We start at looking at the fastest time scale  $l = M$ . We have that the iterates  $\theta_{\alpha^M}^{i,k}$  admit limit processes  $\theta_{\alpha^M}^i$  defined by (6.10)

$$\begin{aligned} (6.11) \quad & \dot{\theta}_{\alpha^M}^i = 0, \quad i < M \\ & \dot{\theta}_{\alpha^M}^M = \bar{F}^M(\theta_{\alpha^M}) = \bar{F}^M(\theta_{\alpha^M}^1, \dots, \theta_{\alpha^M}^M) + b_{\alpha^M}^M \end{aligned}$$

By A 5.6 (ODE solutions), for any  $\theta^{<M}$ , solutions of (6.11) converge to the point  $(\theta^{<M}, \zeta^M(\theta^{<M}))$ . ■

Then at  $(M-1)$ -th timescale, (6.10) becomes

$$\begin{aligned} (6.12) \quad & \dot{\theta}_{\alpha^{M-1}}^i = 0, \quad i < M-1 \\ & \dot{\theta}_{\alpha^{M-1}}^{M-1} = \bar{F}^{M-1}(\theta_{\alpha^{M-1}}^{<M-1}, \zeta^M(\theta_{\alpha^{M-1}}^{<M-1})) + b_{\alpha^{M-1}}^{M-1} \end{aligned}$$

with  $\theta_{\alpha^{M-1}}^{<M-1} = (\theta_{\alpha^{M-1}}^1, \dots, \theta_{\alpha^{M-1}}^{M-1})$ . By A 5.6 (ODE solutions), the solutions of (6.12) converge to the point  $(\theta_{\alpha^{M-1}}^{<M-1}, \zeta^{M-1}(\theta_{\alpha^{M-1}}^{<M-1}))$  for any  $\theta^{<M-1}$ . Repeating this procedure until 2-th timescale, we get

$$(6.13) \quad \dot{\theta}_{\alpha^2}^1 = 0, \quad \dot{\theta}_{\alpha^2}^2 = \bar{F}^2(\theta_{\alpha^2}^{<3}, Z^{\geq 3}(\theta_{\alpha^2}^{<3})) + b_{\alpha^2}^2.$$

Recall that for  $r = 2, \dots, M-1$ ,  $Z^{\geq r}(\cdot)$  is defined by (5.3). By A 5.6 (ODE solutions), for any  $\theta^{<2}$ , the solutions of (6.13) converge to the point  $(\theta^{<2}, \zeta^2(\theta^{<2}))$ . Finally on the slowest timescale, the limit process of  $\theta_{\alpha^1}^{1,k}$  follows

$$\dot{\theta}_{\alpha^1}^1 = \bar{F}^1(\theta_{\alpha^1}^1, Z^{\geq 2}(\theta_{\alpha^1}^1)) + b_{\alpha^1}^1.$$

## 6.2. Proof of Theorem 5.2 Extension to the asynchronous case. Proof: Extension to

the asynchronous case

To switch from the iterate time sequence  $k = 0, 1, 2, \dots$  to the real time sequence, we introduce two sets of functions:

- The  $\Gamma_{c,\alpha^j}^{i,v}(\cdot)$  functions are the piecewise shifted interpolations of the  $\{\Gamma_{k,c}^{ij,v}\}$  sequences at the  $j$  timescale. This means that for a given time  $t$  since  $v$  in iterate time, they return the amount of (scaled) real time elapsed up from  $v$  to the last update.

$$\Gamma_{c,\alpha^j}^{i,v}(t) = \Gamma_{k,c}^{ij,v}, \quad t \in [t_{k,c}^{ij,v}, t_{k+1,c}^{ij,v})$$

- The functions  $N_{c,\alpha^j}^i(\cdot)$  take a given scaled real time  $t$ , and return the iterate time at the  $j$  timescale of the last update at  $i, c$  before  $t$ . That is,

$$N_{c,\alpha^j}^{i,v}(t) = t_{k,c}^{j,v}, \quad t \in [\Gamma_{k,c}^{ij,v}, \Gamma_{k+1,c}^{ij,v})$$

Per construction they are the inverse of the previous  $T_c^i(\cdot)$  functions, in the sense that

$$(6.14) \quad N_{c,\alpha^j}^{i,v}(\Gamma_{c,\alpha^j}^{i,v}(t)) = t_{k,c}^{j,v} \text{ on } [t_{k,c}^{ij,v}, t_{k+1,c}^{ij,v})$$

There exists a straightforward relationship between the continuous interpolations at the real and the  $M$  different iterate timescales, with just a time change:

$$(6.15) \quad \hat{\theta}_{c,\alpha^j}^{i,v}(t) = \theta_{c,\alpha^j}^{i,v}(N_{c,\alpha^j}^{i,v}(t)), \quad \theta_{c,\alpha^j}^{i,v}(t) = \hat{\theta}_{c,\alpha^j}^{i,v}(\Gamma_{c,\alpha^j}^{i,v}(t))$$

*Proof.* Like in Theorem 5.1, we can look at the behavior of the shifted interpolates at every timescale  $l = \{2, \dots, M\}$ , for any real time  $v > 0$ :

$$\theta_{p_c^i(v)+k+1,c}^i = \theta_{p_c^i(v)+k,c}^i + \alpha_k^l \frac{\alpha_k^i}{\alpha_l^l} (F^i(\hat{\theta}_{c,\alpha^j}^{i,v}(\Gamma_{k,c}^{il,v}), \xi_{k,c}^i) + \delta U_{k,c}^i + B_{k,c}^i),$$

Note that  $\Gamma_{k,c}^{il,v} = \Gamma_{c,\alpha^j}^{i,v}(t_{k,c}^{ij,v})$ , and here  $F_k^i$  must be written as a function of the real time interpolation, since not all elements in  $\theta$  are at their  $k$ th update. Then, we can write the shifted continuous interpolations of (5.6) at the  $l$  timescale:

$$(6.16) \quad \theta_{c,\alpha^l}^{i,v}(t) = \theta_{p_c^i(v),c}^i + \bar{G}_{c,\alpha^l}^{i,v}(t) + M_{c,\alpha^l}^{i,v}(t) + H_{c,\alpha^l}^{i,v}(t) + A_{c,\alpha^l}^{i,v}(t) + B_{c,\alpha^l}^{i,v}(t)$$

where  $m^{(i,l)}(t) = \left\{ k \mid t_{k,c}^{il,v} \leq t < t_{k+1,c}^{il,v} \right\}$  and

$$\bar{G}_{c,\alpha^l}^{i,v}(t) = \sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_j^l \frac{\alpha_n^i}{\alpha_n^l} \bar{F}_n^i(\hat{\theta}^i(\Gamma_{n,c}^{il,v})), \quad M_{c,\alpha^l}^{i,v}(t) = \sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} \delta U_{n,c}^i$$

$$B_{c,\alpha^l}^{i,v}(t) = \sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} \delta B_{n,c}^i, \quad H_{c,\alpha^l}^{i,v}(t) = \sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} \left[ F_n^i(\hat{\theta}^i(\Gamma_{n,c}^{il,v}), \xi_{n,c}^i) - \bar{F}^i(\hat{\theta}^i(\Gamma_{n,c}^{il,v})) \right].$$

1324 and  $A_{\alpha^l}^{i,v}(t) = 0$  if  $i \neq l < M$  or  $l = M$ ,

$$1325 \quad (6.17) \quad A_{c,\alpha^l}^{i,v}(t) = \sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_n^l \left[ \bar{F}^l(\hat{\theta}_c^i(\Gamma_{n,c}^{il,v})) - \bar{F}^l(\hat{\theta}_c^{<l+1}(\Gamma_{n,c}^{il,v}), Z^l(\hat{\theta}_c^{<l}(\Gamma_{n,c}^{il,v}))) \right]$$

1326  $\Gamma_{c,\alpha^l}^{i,v}$  admits a similar decomposition:

$$1327 \quad (6.18) \quad \begin{aligned} \Gamma_{c,\alpha^l}^{i,v}(t) &= \Gamma_{p_c^i(v),c}^{il,0} + \sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} \tau_{n,c}^i \\ &= \underbrace{\Gamma_{p_c^i(v),c}^{il,0} + \sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} \mathbb{E}_{k,c}^{i,\tau}[\tau_{n,c}^i]}_{G\Gamma_{c,\alpha^l}^{i,v}} + \underbrace{\sum_{n=p_c^i(v)}^{m^{(i,l)}(t)-1} \alpha_n^l \frac{\alpha_n^i}{\alpha_n^l} (\tau_{n,c}^i - \mathbb{E}_{k,c}^{i,\tau}[\tau_{n,c}^i])}_{M\Gamma_{c,\alpha^l}^{i,v}} \end{aligned}$$

1328 for which we have introduced the two continuous functions  $G\Gamma_{c,\alpha^l}^{i,v}$  and  $M\Gamma_{c,\alpha^l}^{i,v}$ . Note that  
1329  $M\Gamma_{c,\alpha^l}^{i,v}$ , like  $M_{c,\alpha^l}^{i,v}$ , returns a sum of martingale differences.

1330 As in section ??, we use the uniform integrability of the  $Y_{k,c}^i$  (A 5.1) and apply Lemma 6.1  
1331 to the element-wise iterates  $\theta_{k,c}^i$  to derive the tightness of the interpolation processes  $\{\theta_{c,\alpha^l}^{i,k}, B_{c,\alpha^l}^{i,k}\}$ ,  
1332 as well as the Lipschitz continuity of the paths of any of their weak sense limit processes. The  
1333 same reasoning, applied to the uniform integrability of the  $\tau_{k,c}^i$  (A 5.7), gives us the tightness  
1334 of the  $\Gamma_{c,\alpha^l}^{i,v}$ . Note that owing to A 5.7  $\mathbb{E}_{k,c}^{i,\tau}[\tau_{k,c}^i] = u_c^i(\hat{\theta}^i(T_{k,c}^i))$  and the fact that  $u_c^i$  is strictly  
1335 positive, then since  $t$ ,  $N_{c,\alpha^l}^{i,v}(\Gamma_{c,\alpha^l}^{i,v}(t)) = t$  ((6.14)) with  $N_{c,\alpha^l}^{i,v}, \Gamma_{c,\alpha^l}^{i,v}$  continuous, we conclude  
1336 that  $N_{c,\alpha^l}^{i,v}$  is also tight. Finally, from this latter fact and the tightness of  $\theta_{c,\alpha^l}^{i,v}$  we conclude  
1337 the tightness of  $\hat{\theta}_{c,\alpha^l}^{i,v}(\cdot) = \theta_{c,\alpha^l}^{i,v}(N_{c,\alpha^l}^{i,v}(\cdot))$ .

Writing  $\{v_k\}_k > 0$  a sequence of real times such that  $v_k \rightarrow \infty$  as  $k \rightarrow \infty$ , we extract  
a subsequence of  $\{\theta_{\alpha^l}^{i,k}, B_{\alpha^l}^{i,k}, \Gamma_{c,\alpha^l}^{i,v_k}\}$ , a further subsequence indexed by  $(n_k^i)$  as  $k \rightarrow \infty$  and  
processes  $\{\theta_{\alpha^l}^i, B_{\alpha^l}^i, \Gamma_{c,\alpha^l}^i, N_{c,\alpha^l}^{i,v}, \hat{\theta}_{\alpha^l}^i\}$  with Lipschitz continuous paths such that uniformly

$$(\theta_{c,\alpha^l}^{i,v_{n_k^i}}, B_{c,\alpha^l}^{i,v_{n_k^i}}, \Gamma_{c,\alpha^l}^{i,v_{n_k^i}}, N_{c,\alpha^l}^{i,v_{n_k^i}}, \hat{\theta}_{c,\alpha^l}^{i,v_{n_k^i}}) \rightarrow (\theta_{c,\alpha^l}^i, B_{c,\alpha^l}^i, \Gamma_{c,\alpha^l}^i, N_{c,\alpha^l}^i, \hat{\theta}_{\alpha^l}^i)$$

1338 First, passing respectively (6.15) and (6.14) to the weak limit, we have

$$1339 \quad (6.19) \quad \hat{\theta}_{c,\alpha^l}^i(t) = \theta_{c,\alpha^l}^i(N_{c,\alpha^l}^i(t)) \quad N_{c,\alpha^l}^i(\Gamma_{c,\alpha^l}^i(t)) = t$$

1340 The characterization of the limit processes  $\Gamma_{c,\alpha^l}^i$  and  $\hat{\theta}_{c,\alpha^l}^i$  can be done following the same  
1341 reasoning as before for (6.4)-(6.7). First to characterize  $\hat{\theta}_{c,\alpha^l}^i$ , we recall that per A 5.1 the  
1342 implicitly defined component-wise  $\delta U_{k,c}^i$  are martingale differences, the noise processes  $\xi_{k,c}^i$   
1343 satisfy A 5.2 and the averaging condition A 5.4. We conclude:

$$(6.20) \quad \theta_{c,\alpha^l}^i(t) = \theta_{c,\alpha^l}^i(0) + \int_0^t \bar{g}_{\alpha^l}^i(s) ds + B_{\alpha^l}^i(t)$$

1345

$$(6.21) \quad \bar{g}_{\alpha^l}^i(\cdot) = \lim_{(n,v_n) \rightarrow \infty} \frac{\alpha_n^i}{\alpha_n^l} F^i(\hat{\theta}_{c,\alpha^l}^{i,v}(\Gamma_{c,\alpha^l}^{i,v}(\cdot))) = \begin{cases} 0, & i < l, \\ \bar{F}^i(\hat{\theta}_{c,\alpha^l}^i(\Gamma_{c,\alpha^l}^i(\cdot))), & i = l \end{cases}$$

1347 The same reasoning is applied to  $\Gamma_{c,\alpha^l}^i$ . Here we only need to handle the martingale error  
 1348  $M\Gamma_{c,\alpha^l}^{i,v}$ , so that the characterization of the limit process follows immediately:

$$(6.22) \quad \Gamma_{c,\alpha^l}^i(t) = \Gamma_{c,\alpha^l}^i(0) + \int_0^t u_c^i(\hat{\theta}_{\alpha^l}^i(\Gamma_{c,\alpha^l}^i(s))) ds$$

1350 From (6.20) and (6.19), it follows that for  $i = l$ :

$$\begin{aligned} \dot{\hat{\theta}}_{c,\alpha^l}^i(t) &= \dot{\theta}_{c,\alpha^l}^i(N_{c,\alpha^l}^i(t)) \dot{N}_{c,\alpha^l}^i(t) \\ &= \left[ \bar{F}^i(\hat{\theta}_{c,\alpha^l}^i(\Gamma_{c,\alpha^l}^i(N_{c,\alpha^l}^i(t)))) + B_{\alpha^l}^i(t) \right] \dot{N}_{c,\alpha^l}^i(t) \end{aligned}$$

1352 It remains to characterize  $\dot{N}_{c,\alpha^l}^i(\cdot)$ . From (6.19)  $N_{c,\alpha^l}^i(\Gamma_{c,\alpha^l}^i(t)) = t$ , we have

$$\dot{N}_{c,\alpha^l}^i(\Gamma_{c,\alpha^l}^i(t)) = \frac{1}{\dot{\Gamma}_{c,\alpha^l}^i(t)} = \frac{1}{u_c^i(\hat{\theta}_{\alpha^l}^i(\Gamma_{c,\alpha^l}^i(t)))}$$

1354 We conclude that limit process  $\hat{\theta}_{c,\alpha^l}^i$ , for any component  $c \in \{1, \dots, C\}$  and  $i \leq l$  follow the  
 1355 mean ODE, with  $\hat{B}_{\alpha^l}^i(t)$  the reflection term:

$$(6.23) \quad \begin{aligned} \dot{\hat{\theta}}_{c,\alpha^l}^i &= 0, \quad i < l \\ \dot{\hat{\theta}}_{c,\alpha^l}^l &= \frac{\bar{F}^l(\hat{\theta}_{c,\alpha^l}^l)}{u_c^l(\hat{\theta}_{c,\alpha^l}^l)} + \hat{B}_{\alpha^l}^l(t) \end{aligned}$$

1357 Then, as before, we use A 5.9 (ODE solutions) and proceed from the fastest scale  $M$  up  
 1358 to the slowest timescale 1, at which the limit processes of  $\hat{\theta}_{c,\alpha^1}^1$  for any  $c \in \{1, \dots, M\}$ , finally  
 1359 follows

$$\dot{\hat{\theta}}_{c,\alpha^1}^1 = \frac{\bar{F}^1(\theta_{c,\alpha^1}^1, Z^{\geq 2}(\theta_{c,\alpha^1}^1))}{u_c^1(\hat{\theta}_{c,\alpha^1}^1)} + \hat{B}_{c,\alpha^1}^1(t).$$

■